

Production Networks and R&D Allocation

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Abstract

This paper investigates how production networks shape firms' R&D decisions and the resulting aggregate inefficiencies. We develop a dynamic model in which firms form supply chains through a persistent matching process and leverage those links to produce and sell new varieties. The model features two sources of R&D misallocation. First, market power distortions reduce private returns to innovation, particularly for older firms in long supply chains. Second, a positive externality arises from love of variety operating through firm-specific relationships: when a firm innovates, its trading partners gain access to a richer menu and produce more efficiently, yet they pay only for quantities purchased, not for variety access. Private R&D therefore falls short of the social optimum, with underinvestment most severe for young firms rapidly expanding their networks. We estimate the model using Japanese firm-to-firm transaction and patent data. The first-best allocation lies close to the decentralized outcome because the two sources of misallocation push in opposite directions: correcting market power tilts R&D toward older firms, while internalizing the externality tilts it toward younger firms. These forces largely offset, so the welfare gain comes primarily from correcting market power, not from reallocating R&D.

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1 Introduction

Firms' research and development (R&D) decisions are fundamentally shaped by their position within production networks. Consider Intel developing a new processor. Intel does not innovate in isolation but leverages an established network of equipment suppliers and silicon foundries. At the same time, Intel's customers, PC manufacturers like Dell, HP, and Lenovo, gain access to this new chip as an input for their own products. This interconnectedness means that a firm's R&D choices are intrinsically linked to its trading partners. How do production networks shape R&D decisions? What inefficiencies arise, and how large are they?

While the literature has extensively studied how innovation spreads through knowledge networks (Liu and Ma, 2023), the channel through which the *production network itself* shapes R&D incentives remains less explored. Understanding this channel and the inefficiencies that may arise is crucial for optimal innovation policy. We develop a model that captures the interplay between production networks and R&D decisions, and we quantify the resulting welfare effects.

Motivated by empirical patterns from a unique Japanese dataset, we build a dynamic model with three core features. First, firms build networks of suppliers over their life cycles through an exogenous matching process that combines random matching with relationship persistence. This process generates an age-dependent network structure where older firms accumulate larger and more valuable supplier networks. Second, firms invest in R&D to create new product varieties. Third, and most critically, successful innovation leverages existing network connections: new products are produced using established suppliers and sold through existing customer relationships.

Despite these features, the firm's dynamic problem remains tractable. Dynamic decision-making with sticky buyer and supplier relationships is typically difficult because a firm must track each trading partner and that partner's network, causing the state space to explode. We preserve tractability by assuming that each firm is randomly matched with a continuum of suppliers and customers whose total mass grows with firm age. This assumption does not pin down a deterministic age-size profile. Even when innovation stochastically adds and removes product lines, the model remains tractable as long as product-level separability, in the spirit of Klette and Kortum (2004), is maintained.

This framework reveals two sources of R&D misallocation. The first stems from standard market power distortions: markups and double marginalization reduce private returns to innovation, particularly for older firms embedded in long supply chains. The second represents our novel contribution: a *network-formation externality* arising from love of variety

operating through firm-specific relationships. Unlike standard models where new varieties benefit the entire market, in our framework only the innovator's trading partners gain access to new products. Crucially, these two sources push R&D allocation in opposite directions across the firm age distribution.

Return to the Intel example. Suppose Intel already offers Core i3 and Core i5 processors. When Intel develops the Core i7, it becomes a more attractive supplier, one with a richer product line. Crucially, only Intel's existing and future customers can access this new chip, not the economy at large. Those customers value the richer menu: more chip varieties allow PC manufacturers like Dell and HP to better match their product specifications, improving production efficiency. This is the love of variety effect under CES production.

Yet the value of a richer menu goes uncompensated: customers pay for chips purchased, not for access to more options. This is a positive externality. Intel's innovation benefits its customers' production efficiency, but Intel captures none of this value. Private R&D investment therefore falls short of the social optimum.

The magnitude of this externality depends on network dynamics. Young firms are rapidly expanding their networks, generating large uncompensated spillovers. Old firms have mature networks where the externality is smaller. This age-dependence is central to our results.

The externality arises at two margins. Existing customers automatically gain access to the new product, but do not pay extra for this expanded menu. Future customers will also benefit when they begin trading with the innovator, yet no payment occurs when the relationship forms. In both cases, the innovator fails to capture the value of variety access.

Our quantitative analysis shows that the social planner's optimal R&D allocation is nearly identical to the decentralized equilibrium. Why? The two sources of misallocation largely offset. Market power distortions suppress returns to innovation more severely for older firms embedded in longer supply chains; correcting this pushes R&D toward older firms. The network-formation externality, by contrast, is largest for young firms rapidly expanding their networks; internalizing it pushes R&D toward younger firms. These age-dependent effects work in opposite directions and nearly cancel. As a result, the welfare gain comes primarily from correcting market power in goods markets, not from reallocating R&D.

This paper connects the literature on the macroeconomic importance of networks, typified by Acemoglu and Carvalho (2012), with the literature on endogenous growth, such as Romer (1990) and Klette and Kortum (2004). Our primary contribution is to integrate these two strands, showing how the transactional structure of a production network shapes R&D incentives and creates a novel set of allocative inefficiencies.

The production networks literature has demonstrated that network structure serves as a potent amplifier of economic shocks while creating misallocation in factor markets. Research by Liu (2019), Baqaee and Farhi (2020), Bigio and La'O (2020), Baqaee et al. (2023), and Osotimehin and Popov (2023) has shown how market power and markup heterogeneity, transmitted through input-output linkages, generate substantial aggregate welfare losses. While our analysis of market power distortions builds directly on these insights, we extend this mechanism beyond static labor allocation to encompass the allocation of R&D resources.

A growing body of work examines the endogenous evolution of these networks, focusing on link formation through partner search and matching to explain the economy's structure (Oberfield, 2018; Arkolakis et al., 2023; Eaton et al., 2022; Huneeus, 2018; Bernard et al., 2022; Acemoglu and Azar, 2020). Our model shares this dynamic perspective, with network connections evolving as firms mature, a pattern also explored by Aekka and Gaurav (2023) and Asai and Nirei (2025). Our contribution is to shift the central margin from partner search to R&D investment, thereby identifying a previously unrecognized externality: the network-formation externality of innovation.

This externality provides a new perspective on R&D misallocation, distinct from inefficiencies arising through "knowledge networks" studied by Liu and Ma (2023), Cai and Li (2019), and Cai and Tian (2021). Our mechanism operates through the production network: the inefficiency stems not from imitation but from innovators' inability to capture the love of variety benefits they create for their buyers. This differs from other R&D misallocation sources, including crowding out (Acemoglu et al., 2018) or innovation spillovers (Aghion et al., 2023; Ayerst, 2023). The interaction between our externality and market power distortions produces our main result: two opposing forces that largely offset each other.¹

The rest of the paper is organized as follows. Section 2 presents the data and empirical findings on the relationship between production networks, firm age, and R&D, including reduced-form evidence that innovation propagates through supply chains. Section 3 develops a dynamic model of production networks and R&D based on these empirical findings. Section 4 analyzes the misallocation of R&D workers in the decentralized equilibrium compared to the social planner's solution, and provides a micro-foundation linking the model to the reduced-form evidence. Section 5 presents a quantitative exercise, estimating the model parameters and comparing the model's predictions with the data.

¹Empirical work using inter-firm network data (Boehm et al., 2019; Bernard et al., 2022, 2019; Carvalho et al., 2020; Bai et al., 2023; Daisuke, 2017) has focused on short-run shock propagation. Our dataset, combining firm-level linkages with patent records, provides the first evidence on long-term supply chain and innovation dynamics.

Finally, Section 6 concludes the paper.

2 Motivating Facts on Production Networks and Innovation

Before developing the model, we establish empirical facts that discipline our assumptions. The first two facts characterize how production networks evolve with firm age; they pin down the matching process. The third fact shows that innovation propagates through supply chains; it supports our key assumption that new products inherit the innovating firm’s existing trading relationships (Assumption 1 in Section 3).

Specifically: (i) older firms have more trading partners, (ii) older firms trade disproportionately with other older firms, and (iii) a firm’s R&D increases when its trading partners innovate. We document these facts using Japanese firm-level data on B2B transactions and patents.

2.1 Data Sources

Our analysis relies on three main datasets: the TDB Center for Advanced Empirical Research on Enterprise and Economy (TDB-CAREE) archive, the IIP Patent Database, and the OECD Triadic Patent Families (TPF) database.

We obtained firm-level data from TDB-CAREE at Hitotsubashi University. The data originate from credit research reports compiled by Teikoku Databank (TDB), a major private credit research company. The dataset contains firm characteristics such as firm age, industry classification (based on codes that closely match the Japan Standard Industrial Classification), location of headquarters, number of employees, and sales revenue. Crucially for our analysis, each firm also reports the identities of its major suppliers and customers, allowing us to construct the production network of supplier-customer linkages. Because TDB-CAREE curates the full TDB universe, the coverage extends beyond publicly listed firms and therefore surpasses sources such as Compustat.

The second dataset is the IIP Patent Database, developed for patent statistical analysis using standardized data from the Patent Office. As the IIP Patent Database lacks corporate information, it is linked with the NISTEP database, which connects patent data to company names and identification numbers. Identification numbers from NISTEP then allow us to merge the IIP Patent Database with the TDB. For firms without identification numbers, we integrate the databases as much as possible using addresses and names.

The third dataset is the OECD Triadic Patent Families (TPF) database, which combines patent applications filed with the European Patent Office (EPO), the Japan Patent Office

(JPO), and the United States Patent and Trademark Office (USPTO) into patent families based on common priority applications. The primary data source for the TPF is the EPO's Worldwide Patent Statistics Database, which provides harmonized and comparable information on patents from the EPO, JPO, and USPTO. We use information on the IPC (four digits), the year of application (earliest filing date at the Japan Patent Office), and the nationality of the applicant for each patent.

2.2 Network Patterns by Firm Age

We begin with the first two facts, which discipline the matching process in our model.

Fact 1: The number of trading partners increases with firm age, but the rate of increase diminishes over time.

Fact 2: Older firms trade disproportionately with other older firms, mainly because trading relationships are sticky.

We now document each fact in turn.

2.2.1 Age-dependent network relationships

We begin by analyzing the cross-sectional relationship between firm age and the number of trading partners. Figure 1a shows local linear regressions of the numbers of suppliers and buyers on firm age. Both series increase monotonically with age. After a steep increase in the first half of the life cycle, the pattern transitions to a more gradual, roughly log-linear rise. Figure 1b plots the corresponding growth rates, confirming that partner growth is high early in the life cycle and converges to a constant value. The number of suppliers tends to slightly exceed the number of buyers, but the slopes are not markedly different.

2.2.2 Positive age assortative matching by age in cross-section

Next, we turn to the features of trading partners conditional on age. Figure 2a plots trading partners' ages against firms' own ages. Older firms tend to match with older partners, indicating positive assortative matching by age in the cross-section.

To understand this assortative matching, we consider existing links (stocks) and the formation or withdrawal of business relationships (flows). We first examine the age heterogeneity in current-period matches within the flows. Figure 2b plots a local linear regression of partners' ages on firms' own ages. The flows exhibit only one to two years of heterogeneity, considerably smaller than the heterogeneity in the cross-section. Next, to examine the age dependence of link termination, Figure 3 plots the survival of each link as

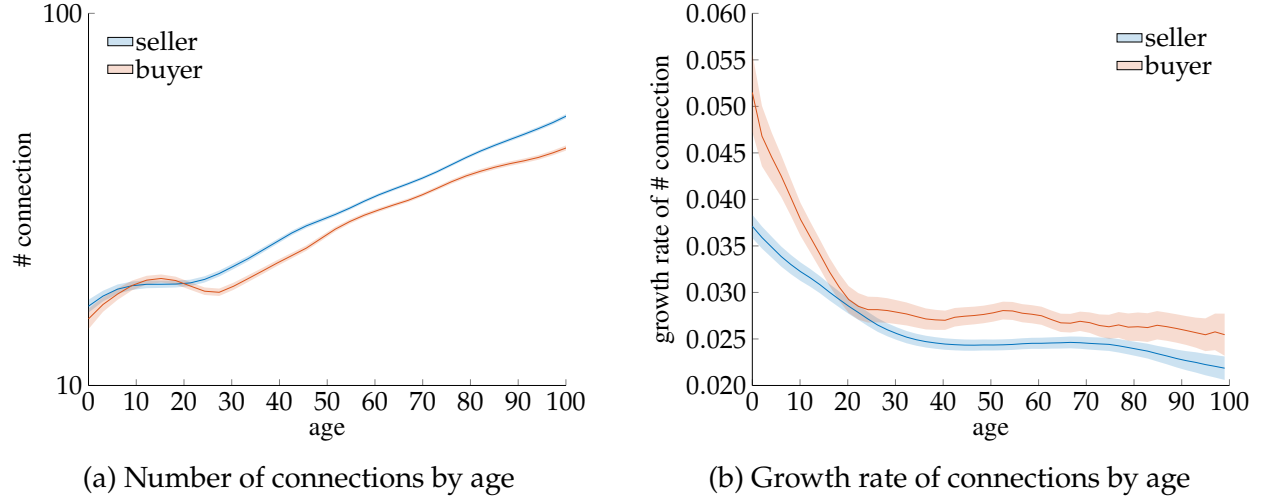


Figure 1: Lifecycle of production networks.

Notes: The figure shows local linear regression estimates relating firms' connections to firm age with controls. The sample covers 1998–2019. The control terms include prefecture, industry, and year fixed effects. The two lines represent the supplier and buyer sides as separate regressions. Shaded areas indicate the 95 percent confidence intervals.

a function of the number of trading years between firms for each supplier and buyer age quartile group. The probability that a link breaks decreases with the number of transaction years, but there is little significant heterogeneity across seller and buyer age groups.

Why does age assortativity arise, even though the formation of new links is independent of firm age? The reason is that when business relationships are sticky, firms and their past counterparts age together. According to Figure 3, it takes more than 10 years for 90% of business relationships to dissolve. This high degree of stickiness generates the observed positive age assortativity of matching.

Finally, Figure 4 presents local linear regressions relating R&D activity to firm age. Both proxy measures of R&D activity increase monotonically with age.

2.3 Innovation Propagation through Supply Chains

We now turn to the third fact, which disciplines our key modeling assumption.

Fact 3: A firm's R&D increases when its trading partners innovate.

This fact supports Assumption 1 in our model: new products inherit the innovating firm's existing trading relationships. When a supplier innovates, its buyers automatically gain access to additional varieties, which raises their efficiency and R&D incentives. We test whether this propagation occurs in the data using instrumental variables.

We use an integrated database that combines firm-level transaction data from TDB with

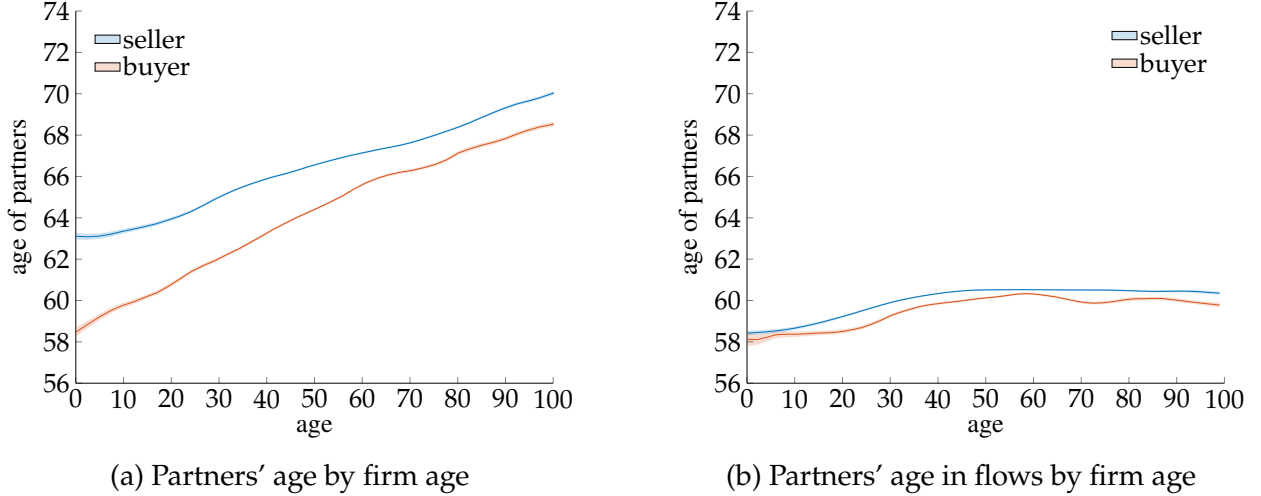


Figure 2: Positive age assortativity.

Notes: The figure shows local linear regression estimates relating trading partners' ages to firm age with controls. The sample covers 1998–2019. The control terms include prefecture, industry, and year fixed effects. The two lines represent the supplier and buyer sides as separate regressions. Shaded areas indicate the 95 percent confidence intervals.

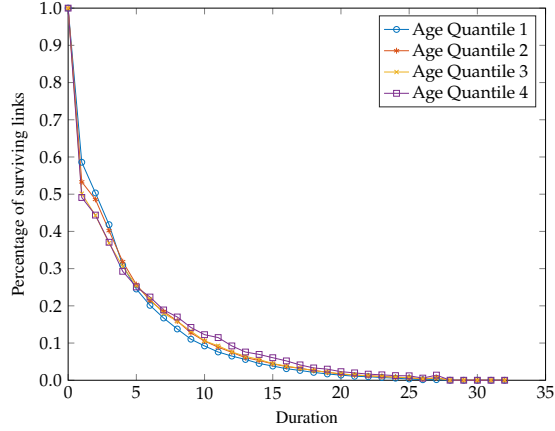
patent information from the IIP Patent Database maintained by the JPO.

While our formulation builds on Acemoglu et al. (2015) and Liu and Ma (2023), it differs in two key respects. First, instead of aggregated industry- or technology-class variables, we exploit firm-level transaction links alongside patent citation relationships. Second, whereas Liu and Ma (2023) document technological spillovers and Acemoglu et al. (2015) study foreign patent shocks transmitted through production networks, our data allow us to separate transactional and technological ties at the firm level. Appendix Section A.1 documents that the two networks still overlap, even when observed at the firm level.

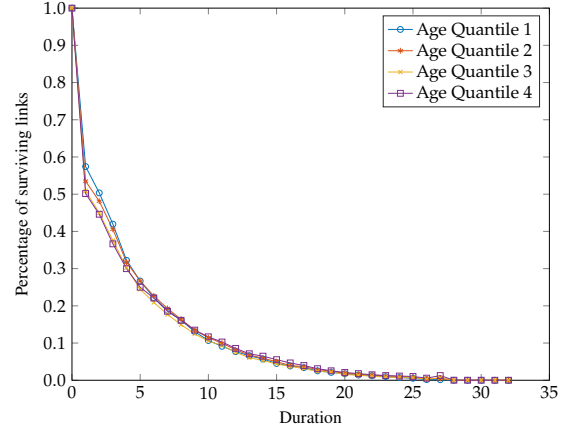
We estimate regressions on firm-level data from 1994 to 2019, measuring each firm's R&D activity by the number of patent applications and identifying production-network connections through the TDB database. Our baseline specification is

$$\log |\text{Patents}|_{i,t} = \beta_1 \log \frac{|\text{Sellers' Patents}|_{i,t}}{|\text{Sellers}|_{i,t}} + \beta_2 \log \frac{|\text{Buyers' Patents}|_{i,t}}{|\text{Buyers}|_{i,t}} + \text{controls}_{i,t} + \varepsilon_{i,t},$$

where the controls include firm and year fixed effects. Here, $|\text{Sellers' Patents}|_{i,t}$ ($|\text{Buyers' Patents}|_{i,t}$) denotes the total number of patent applications filed in year t by all sellers (buyers) of firm i . The ratios $|\text{Sellers' Patents}|_{i,t} / |\text{Sellers}|_{i,t}$ and $|\text{Buyers' Patents}|_{i,t} / |\text{Buyers}|_{i,t}$ therefore capture the average R&D activity of a firm's trading partners. To mitigate endogeneity



(a) Survival links by seller's age



(b) Survival links by buyer's age

Figure 3: Survival links.

Notes: This figure plots the duration of business relationships between firms. To mitigate bias from sample observability, the sample is restricted to connections where (1) the connection is first observed more than one year after the period in which both firms were first observed and (2) the connection is no longer observed more than one year before the period in which either firm was last observed. The figure on the left classifies business relationships by the age quartiles of supplier firms. The right figure classifies business relationships by the age quartiles of buyer firms.

concerns, we construct instrumental variables based on exposure to foreign patents:

$$\text{IV for } \frac{|\text{Sellers' Patents}|_{i,t}}{|\text{Sellers}|_{i,t}} = \sum_c \frac{|\text{Sellers' Patents}|_{i,c,t-1}}{|\text{Sellers' Patents}|_{i,t-1}} \times |\text{Foreign Patents}|_{c,t-1},$$

$$\text{IV for } \frac{|\text{Buyers' Patents}|_{i,t}}{|\text{Buyers}|_{i,t}} = \sum_c \frac{|\text{Buyers' Patents}|_{i,c,t-1}}{|\text{Buyers' Patents}|_{i,t-1}} \times |\text{Foreign Patents}|_{c,t-1}.$$

Here, $|\text{Foreign Patents}|_{c,t-1}$ is the number of triadic patents in class c filed at time $t - 1$ at the Japan Patent Office by foreign firms, after excluding Japanese applicants. Triadic patents capture global technology trends rather than domestic demand shocks, yielding exogenous shifters for Japanese firms. By observing firm-level technology exposure directly, we can construct the above instruments without sector-level aggregation. The weights $|\text{Sellers' Patents}|_{i,c,t-1} / |\text{Sellers' Patents}|_{i,t-1}$ and $|\text{Buyers' Patents}|_{i,c,t-1} / |\text{Buyers' Patents}|_{i,t-1}$ capture the exposure of each firm to patent class c through its sellers and buyers, respectively.

Table 1 reports ordinary least squares (OLS) and instrumental variable (IV) estimates. In both sets of regressions the coefficients on the average number of sellers' and buyers' patent applications are positive and statistically significant, indicating that a firm's R&D activity increases with the innovation intensity of its trading partners. The IV estimates in

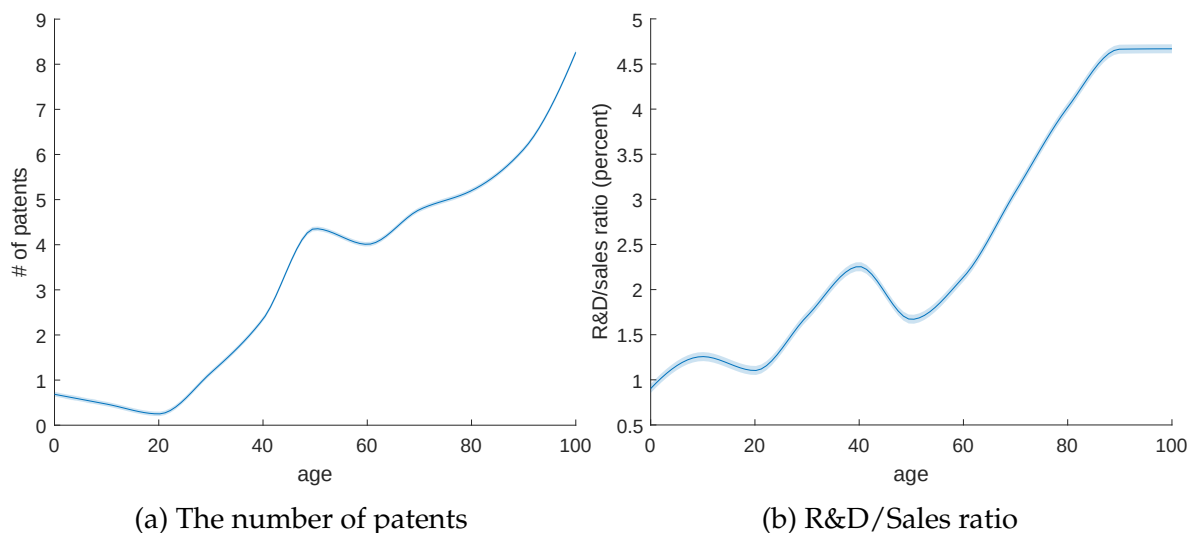


Figure 4: R&D and age.

Notes: The figure shows local linear regression estimates relating R&D measures to firm age with controls. The sample covers 1998–2019. The control terms include prefecture, industry, and year fixed effects for the firm and its clients. Shaded areas indicate the 95 percent confidence intervals.

columns (4)–(6) are larger in magnitude than the corresponding OLS estimates, consistent with downward bias in the naive specification.

To verify that the results stem from production-network mechanisms rather than direct technological spillovers, Table 2 shows that the coefficients remain sizable and significant even when we exclude patents that cite the applicant’s buyers or sellers. This pattern supports the interpretation that production relationships, rather than shared patent classes, drive the observed spillovers.

Table 1: Regression Results

	(1)	(2)	(3)	(4)	(5)	(6)
$\log \frac{ \text{Sellers' Patents} _{i,t}}{ \text{Sellers} _{i,t}}$	0.020*** (0.003)		0.019*** (0.003)	0.068*** (0.018)		0.062*** (0.018)
$\log \frac{ \text{Buyers. Patents} _{i,t}}{ \text{Buyers} _{i,t}}$		0.022*** (0.003)	0.020*** (0.003)		0.077*** (0.021)	0.073*** (0.021)
year fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
firm fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Fstat				840.4	759.8	373.5
Observations	53,553	53,553	53,553	53,553	53,553	53,553

Standard errors in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Notes: Standard errors are clustered at the industry $\times$ year level.

Table 2: Regression Results (excluding patenting relationships)

	(1)	(2)	(3)	(4)	(5)	(6)
$\log \frac{ \text{Sellers' Patents} _{i,t}}{ \text{Sellers} _{i,t}}$	0.018*** (0.003)		0.017*** (0.003)	0.049** (0.019)		0.043** (0.019)
$\log \frac{ \text{Buyers. Patents} _{i,t}}{ \text{Buyers} _{i,t}}$		0.020*** (0.003)	0.019*** (0.003)		0.081*** (0.023)	0.079*** (0.023)
year fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
firm fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Fstat				805.3	690.5	339.0
Observations	51,338	51,338	51,338	51,338	51,338	51,338

Standard errors in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Notes: Standard errors are clustered at the industry $\times$ year level.

In sum, firms innovate more when their trading partners do so, even after addressing endogeneity with instrumental variables. This supports our key modeling assumption: new products inherit the innovating firm's existing trading relationships, so innovation

propagates through supply chains.

We now build a model that captures all three facts.

3 A Model of Production Networks and R&D

We now build a model that captures the three facts from Section 2. Facts 1 and 2 (older firms have more partners; older firms trade with older firms) are captured by a matching process with random link formation and relationship persistence. Fact 3 (innovation propagates through supply chains) is captured by Assumption 1: new products inherit the innovating firm's existing trading relationships.

We build on Klette and Kortum (2004), a canonical model of innovation and firm dynamics, and introduce firm-to-firm matching. The key departure is that successful R&D leverages existing network connections: new products are produced using established suppliers and sold through existing customer relationships.

3.1 Settings

A unit measure of infinitely lived households supplies a unit measure of production workers, R&D workers, and entrepreneurs.² Households have preferences over a final consumption good, $U_0 = \int_0^\infty \exp(-\rho t) \log Y(t) dt$, where $\rho > 0$ is the discount rate and $Y(t)$ is consumption. The budget constraint is $\dot{A}(t) \leq r(t)A(t) + w(t) + w_H(t) + w_E(t) - P(t)Y(t)$ with the standard no-Ponzi condition, where $A(t)$ is the asset position, $r(t)$ is the interest rate, and $w(t)$, $w_H(t)$, and $w_E(t)$ denote wages for each type of worker. We set nominal GDP as the numeraire. The Euler equation then implies $r(t) = \rho$. In what follows, we focus on the stationary equilibrium and drop time subscripts when no confusion arises.

Products, Firms, and Production Networks

There is a continuum of intermediate goods, indexed by $\omega \in \Omega(t)$. The measure of intermediate goods evolves through the creation of new varieties and exit. Each intermediate good ω is produced by a monopolist, and a given firm may own multiple product lines and operate several goods simultaneously. Consider a firm $f \in \mathcal{F}$ that owns product line ω . Let $n(f)$ denote the number of product lines owned by firm f . We drop the firm subscript when no confusion arises. Denote by $\mathcal{S}(\omega) \subset \Omega$ the set of products used as inputs for

²Fixing the supply of skilled labor abstracts from the underinvestment that would arise if other workers could perform R&D or if output were directly convertible into R&D. Recent work on R&D misallocation (Aghion et al., 2023; Liu and Ma, 2023) adopts the same assumption.

good ω . Although buyers can be inferred from the inverse mapping of $\mathcal{S}(\cdot)$, for notational simplicity we write $\mathcal{B}(\cdot) : \Omega \rightarrow \Omega$ for the set of buyers in Ω .

Production Structure Given Networks

Firms use production workers and intermediate inputs for production. Intermediate inputs are imperfect substitutes with a constant elasticity of substitution, $\sigma \geq 1$. Production workers and the composite of intermediate inputs are combined in a Cobb-Douglas aggregator with labor share, β ($0 \leq \beta \leq 1$):

$$x(\omega) = \frac{1}{\beta^\beta (1-\beta)^{1-\beta}} l(\omega)^\beta \left(\int_{\omega' \in \mathcal{S}(\omega)} x(\omega', \omega)^{\frac{\sigma-1}{\sigma}} d\omega' \right)^{\frac{\sigma}{\sigma-1} (1-\beta)}, \quad (1)$$

where $l(\omega)$ is demand for production workers used to produce ω ; $x(\omega', \omega)$ is the demand for product ω' used to produce ω .

A representative household has a CES utility function with an elasticity of substitution σ , which is the same elasticity as that for production:

$$Y = \left(\int_{\omega \in \Omega} y(\omega)^{\frac{\sigma-1}{\sigma}} d\omega \right)^{\frac{\sigma}{\sigma-1}}, \quad (2)$$

where $y(\omega) = Y \left(\frac{p(\omega)}{P} \right)^{-\sigma}$ is the final demand for product ω . For all ω , $x(\omega)$ satisfies the following market clearing condition:

$$x(\omega) = \int_{\omega' \in \mathcal{B}(\omega)} x(\omega, \omega') d\omega' + y(\omega), \quad (3)$$

Under monopolistic competition, suppliers charge a constant markup over their marginal cost:

$$p(\omega', \omega) = p^y(\omega') = \mu c(\omega'), \quad (4)$$

where $p(\omega', \omega)$ is the product price that supplier ω' charges to firm ω , $p^y(\omega')$ is the price that supplier ω' charges to the final good consumer, $c(\omega)$ is the marginal cost of product ω , and $\mu = \frac{\sigma}{\sigma-1}$.

R&D, Matching, Entry, and Exit

Firms conduct R&D to acquire product lines of new variety.³ We make the following assumptions about the set of production networks of sellers and buyers for which new products are available.

Assumption 1. *For a new product line ω' derived from the existing firm's product line ω ,*

$$S(\omega') = S(\omega),$$

$$\mathcal{B}(\omega') = \mathcal{B}(\omega)$$

In words, existing firms can develop new products and sell them to existing buyers by using intermediate goods from suppliers of each product line owned by the existing product line.

The product level cost function follow Klette and Kortum (2004):

$$\tilde{\phi}(x, n) = nw_H\phi(\lambda), \tag{5}$$

where $\phi(\lambda) = \frac{1}{\phi}\lambda^\gamma$ with efficiency parameter $\phi > 0$, λ is a per product innovation rate, and $\gamma > 1$.⁴

Firms are matched randomly with other firms at exogenous rates, ζ . The match is exogenously terminated at the rate δ_M or if one of the firms in either side of the match exits. Firms die at an exogenous rate δ_F . Entrepreneurs have access to a linear entry technology, where each R&D worker generates a flow of λ_E . Entrants start with ζ_0 mass of randomly chosen sellers or buyers.

3.2 Characterization of decentralized Equilibrium

The model involves the problem of tracking trading relationships between continuous products, which is impossible to solve in general, but we show that under our setting the equilibrium conditions are summarized by the following functions over the state space of ages, a .

³Our model is based on Klette and Kortum (2004), but abstracts from productivity differences and growth. So instead of creative destruction, we assume variety creation.

⁴This product level cost function can be micro-founded by a constant-returns-to-scale innovation technology that uses R&D workers and the number of products as inputs. Namely, a firm f hires $l_H(f)$ units of skilled workers to add one more product at the flow rate $\Lambda(f) = n(f)^{1-1/\gamma} (\phi l_H(f))^{1/\gamma}$, where Λ is the firm-level flow rate.

Thereafter, let $F(a)$ be the cumulative density of products with respect to age a , and we will use a instead of ω . Let $N(t)$ be the total measure of products. Let $F(a, t)$ be defined so that the fraction of products which is owned by firms with age less than or equal to a at time t is $F(a, t)/N(t)$. Let $N_f(t)$ denote the total mass of firms. Because the total mass of firms evolves according to $\dot{N}_f(t) = \lambda_E - \delta_F N_f(t)$, in the steady state $N_f = \lambda_E/\delta_F$. We begin with a matching distribution. As we will show later, because the optimal innovation rate per product, λ , depends on age, the matching process satisfies the following differential equation.

Proposition 1. *Law of motion of matching process: The distribution of matched products with age less than a' connected with an age a product is given by*

$$\underbrace{\frac{\partial}{\partial a'} m(a', a) + \frac{\partial}{\partial a} m(a', a)}_{\text{time evolution of the matching distribution}} = \underbrace{-(\delta_M + \delta_F) m(a', a)}_{\text{link destruction}} + \underbrace{\frac{\zeta}{N_f} f(a')}_{\text{random matching}} + \underbrace{\lambda(a') m(a', a) + \frac{\zeta_0}{N_f} \lambda_E \delta(a)}_{\substack{\text{innovation rate} \\ \text{new products created by partners of age } a'}}, \quad (6)$$

where $\lambda(a)$ is the innovation rate of age a firm, subject to the boundary conditions

$$m(a'; 0) = \frac{\zeta_0}{N_f} f(a') \quad (7)$$

The matching distribution evolves over time because of the following reasons. First, the measure of matched products increases when the currently linked firm creates a new variety at rate $\lambda(a')$. Second, the matched products are lost when the link is terminated at the exogenous rate δ_M or when the linked firm exits at the exogenous rate δ_F . Finally, the matched products are added when there is random matching with new firms.

In general, if connections change stochastically, one needs to track changes in connections among countless firms each period. However, from the perspective of connected partners, the randomness disappears due to the law of large numbers. Thus, the distribution can be summarized by age. This formulation opens new possibilities for modeling two-sided production network structures.

Next, armed with a matching distribution as a function of age, we can characterize the equilibrium objects of firms using the demand shifters $D(a)$ and the cost shifters $c(a)$. These functions solve the two fixed-point problems described in the following Lemma.

Lemma 1. *The demand shifters and cost shifters satisfy the following fixed-point equations:*

$$D(a) = (1 - \beta) \mu^{-\sigma} \int \left(\frac{c(a')}{w} \right)^{\frac{\beta}{1-\beta}(\sigma-1)} D(a') m(a', a) da' + \underbrace{\frac{PY}{GDP}}_{GDP} \quad (8)$$

$$c(a) = w^\beta \left(\int (\mu c(a'))^{1-\sigma} m(a', a) da' \right)^{\frac{1-\beta}{1-\sigma}} \quad (9)$$

where $P = \mu \left(\int c(a)^{1-\sigma} dF(a) \right)^{\frac{1}{1-\sigma}}$.

The demand shifter represents the relative size of demand faced by firms at age a relative to the size of the economy, PY , as measured by nominal GDP. The size of demand for intermediate goods is the sum of the portion of aggregate demand $D(a')$ coming from different ages toward a for all a' . The presence of the markup in equation (8) implies that the demand for intermediate goods shrinks as double marginalization is repeated for each intermediate goods transaction. The cost shifters (9) are standard, but the point is that the matching function is a function of a , so the cost function can also be summarized by a .

Using Lemma 1, it is easy to recover the product level equilibrium objects as a function of a . The profit generated by a product with age a can be characterized by

$$\pi(a) = \left(1 - \frac{1}{\mu} \right) \left(\frac{\mu c(a)}{P} \right)^{1-\sigma} D(a)$$

The fact that profit is a function of age suggests that the firm's optimization problem can also be summarized in terms of age. A firm with n product lines and age a maximizes the value $V^F(n, a)$

$$\begin{aligned} r(t)V^F(n, a) = & \underbrace{n\pi(a)}_{\text{profits}} - \underbrace{\delta_F n \{V^F(n, a) - V^F(n-1, a)\}}_{\text{product exit}} + \underbrace{V_a^F(n, a)}_{\text{age effect}} \\ & + \max_{\lambda \geq 0} \left[\underbrace{n\lambda \{V^F(n+1, a) - V^F(n, a)\}}_{\text{expansion of variety}} - \underbrace{nw_H \phi(\lambda)}_{\text{R\&D costs}} \right]. \end{aligned} \quad (10)$$

In words, the first term on the right-hand side is the total static profit. The second term is the change in firm value due to the exogenous withdrawal of one of its product lines. The third term is the change in firm value due to aging. The fourth term is the change in firm

value if a product is added when a new product line arrives with a Poisson arrival rate $n\lambda$. The last term is the R&D cost.

We can show that the value of each firm can be expressed as the sum of the value of the product lines, defined as the net present discounted value of profits from a product line. To show this, guess

$$V^F(n, a) = nV(a)$$

where $V(a)$ is the value of product lines owned by an age a firm, and obtain the following HJB equation for $V(a)$:

$$(\rho + \delta_F) V(a) = \pi(a) + V_a(a) + \max_{\lambda \geq 0} [\lambda V(a) - w_H \phi(\lambda)] \quad (11)$$

Finally, the first order condition yields an optimal innovation rate:

$$\lambda(a) = \left\{ \frac{\phi}{\gamma w_H} V(a) \right\}^{\frac{1}{\gamma-1}}, \quad (12)$$

which confirms λ is also a function of age.

Kolmogorov Forward Equations

Standard arguments establish that the differential equation governing the evolution of the product density, $f(a)$ at steady state takes the following form:

$$0 = -\frac{\partial f(a)}{\partial a} + (\lambda(a) - \delta_F) f(a) + \lambda_E \delta(a) \quad (13)$$

where $\delta(a)$ denotes the Dirac delta function, which is zero everywhere except if $a = 0$, and satisfies $\int \delta(a) da = 1$. The time evolution of the firm distribution $\frac{\partial f(a)}{\partial t} = 0$ is consistent with the sum of the four terms in the right-hand side: The first term captures the increase in firm age, the second term captures the increase in the number of product lines due to the successful R&D by age a firm minus the exogenous exit of product lines, and the third term captures the addition of product lines due to the entry of age 0 firm.

Labor Market Clearing Conditions

The labor market clearing condition can be expressed using the demand shifters,

$$w = \beta \mu^{-\sigma} \int \left(\frac{c(a)}{P} \right)^{1-\sigma} D(a) f(a) da \quad (14)$$

The high skilled labor market clearing condition is

$$L_H = \int \phi(\lambda(a)) f(a) da \quad (15)$$

4 R&D Allocation

The decentralized equilibrium characterized in Section 3 may not be efficient. This section identifies two potential sources of R&D misallocation: market power distortions and a positive network-formation externality.

Our approach compares three allocations:

- **Decentralized Equilibrium (DE):** The baseline where firms optimize privately.
- **First-Best Social Planner (SP):** Eliminates market power distortions *and* internalizes the network-formation externality.
- **Second-Best Constrained Planner (CP):** Internalizes the externality *only*, maintaining market power distortions.

Comparing SP to DE reveals total inefficiency. Comparing CP to DE isolates the contribution of the externality. The difference between SP and CP reveals the contribution of market power distortions. These theoretical benchmarks provide the foundation for our quantitative welfare analysis in Section 5.

4.1 Decentralized Equilibrium

We begin with the decentralized equilibrium as the baseline for comparison.

Proposition 2 (Decentralized Equilibrium R&D Allocation). *The allocation of R&D workers in the decentralized equilibrium satisfies:*

$$l_H(a) = \frac{V(a)^{\frac{\gamma}{\gamma-1}}}{\int V(a)^{\frac{\gamma}{\gamma-1}} f(a) da} \propto V(a)^{\frac{\gamma}{\gamma-1}} \quad (16)$$

where the product value function $V(a)$ solves:

$$\begin{aligned} \rho V(a) = & \underbrace{\pi(a)}_{\text{profit}} \underbrace{-\delta_F V(a)}_{\text{product death}} + \underbrace{V_a(a)}_{\text{age effect}} \\ & + \underbrace{[\lambda(a)V(a) - w_H \phi(\lambda(a))]}_{\text{net value of innovation}} \end{aligned} \quad (17)$$

where revenue is $r(a) = \left(\frac{\mu c(a)}{P}\right)^{1-\sigma} D(a)$ and profits are $\pi(a) = \left(1 - \frac{1}{\mu}\right) r(a)$.

This allocation is distorted due to both markup pricing in the goods market and the failure to internalize innovation externalities through the production network.

4.2 First-Best Social Planner

We now characterize the efficient benchmark. The Social Planner eliminates both sources of inefficiency: market power distortions (by setting prices equal to marginal cost) and the network-formation externality (by internalizing love of variety benefits).

The Social Planner maximizes the discounted utility flow:

$$U_0 = \int_0^\infty e^{-\rho t} \log Y(t) dt \quad (18)$$

subject to the laws of motion for product and match distributions (13) and (6), and the R&D labor constraint (15).

The following Proposition characterizes the optimal allocation. The complete derivation, including how the static conditions emerge from the planner's first-order conditions, is provided in Appendix D.

Proposition 3 (First-Best Social Planner Allocation). *The Social Planner's R&D allocation satisfies:*

$$l_H^{SP}(a) \propto \left(V^{SP}(a)\right)^{\frac{\gamma}{\gamma-1}} \quad (19)$$

where the social value function $V^{SP}(a)$ solves:

$$\begin{aligned} \rho V^{SP}(a) = & r^{SP}(a) - \delta_F V^{SP}(a) + V_a^{SP}(a) + [\lambda^{SP}(a)V^{SP}(a) - w_H^{SP} \phi(\lambda^{SP}(a))] \\ & + \int V^M(a, a') \left\{ \frac{\zeta}{N_f} f(a') - \delta_M m(a, a') \right\} da' \end{aligned} \quad (20)$$

and $V^M(a, a')$ is the social value of a match between a supplier of age a and a buyer of age a' , satisfying:

$$(\rho + \delta_F + \delta_M - \lambda^{SP}(a))V^M(a, a') = V_a^M(a, a') + V_{a'}^M(a, a') + (1 - \beta)r^{SP}(a')\Omega(a, a') \quad (21)$$

where:

- $r^{SP}(a) = \left(\frac{c^{SP}(a)}{p^{SP}}\right)^{1-\sigma} D^{SP}(a)$ is social revenue
- $c^{SP}(a)$ and $D^{SP}(a)$ are cost and demand shifters satisfying the same recursive structure as the decentralized equilibrium (Lemma 1), but without markup μ
- $\Omega(a, a') = \frac{c^{SP}(a)^{1-\sigma}}{\int c^{SP}(a'')^{1-\sigma} m(a'', a') da''}$ is supplier a 's expenditure share in buyer a' 's input bundle
- w_H^{SP} is the shadow wage on R&D labor

The match value $V^M(a, a')$ captures the love of variety benefit: when buyer a' has access to supplier a 's products, buyer a' 's production efficiency improves through CES aggregation. The flow value $(1 - \beta)r^{SP}(a')\Omega(a, a')$ represents this net benefit, depending on the buyer's revenue and the supplier's expenditure share.

4.3 Two Channels of R&D Misallocation

Comparing the decentralized equilibrium (Proposition 2) to the social planner (Proposition 3) reveals two sources of inefficiency. This is the core theoretical contribution of the paper.

Distortions from Market Power: The first channel arises from markup pricing in the goods market. While the markup μ is uniform across firms, the network structure creates *heterogeneous degrees of double marginalization*. Older, more central firms are embedded in longer supply chains, causing their revenues, $r(a)$, to be disproportionately suppressed relative to the social planner's equivalent, $r^{SP}(a)$. This distorts the private value of innovation, $V(a)$, and consequently the allocation of R&D resources. This force, related to the static misallocation literature, systematically disincentivizes R&D in older firms.

Network-Formation Externality: The second channel is a positive externality arising from love of variety operating through firm-specific relationships. When a firm creates a new product through R&D, that product becomes an additional variety available only to the firm's buyers, not to the market at large. Under CES production, buyers' efficiency increases with the number of input varieties they can access, the love of variety effect. However, market prices compensate suppliers only for units sold, not for the production efficiency

gains that buyers derive from having access to an additional variety. The innovating firm thus fails to internalize the love of variety benefits it creates for its trading partners. This uninternalized value is quantified as the "network-formation externality wedge":

$$\text{wedge}(a) = \int V^M(a, a') \left\{ \frac{\zeta}{N_f} f(a') - \delta_M m(a, a') \right\} da' \quad (22)$$

This wedge measures the social value of match formation and destruction flows associated with product a as a supplier. The term $V^M(a, a')$ is the present value of love of variety benefits that buyer a' derives from accessing supplier a 's products. The innovating firm does not internalize this value because: (i) prices compensate for purchases, not for variety access; (ii) existing buyers automatically inherit access to new products without renegotiation (Assumption 1); and (iii) future buyers cannot bargain with firms they have not yet matched with. Because this is a positive externality, private R&D investment falls short of the social optimum. The magnitude varies with firm age a : young firms generate larger uncompensated spillovers, so the underinvestment is most severe for them. A quantitative assessment of the resulting misallocation is conducted in Section 5.

4.4 Empirical Support: Strategic Complementarity

The previous subsection identified the network-formation externality theoretically. We now provide a testable prediction that connects the theory to the IV evidence in Section 2.3.

The externality rests on a key mechanism: innovation by one firm affects the R&D incentives of its trading partners. The following proposition formalizes this.

Proposition 4 (Strategic Complementarity in Innovation). *Consider buyer a' in partial equilibrium. Fix the aggregate price index P , R&D wage w_H , suppliers' cost shifters $\{c(a)\}_{a \geq 0}$, buyer's demand shifter $D(a')$, and continuation value $V_{a'}(a')$.*

An increase in the match measure $m(a, a')$ with any supplier a raises the buyer's optimal innovation rate $\lambda(a')$.

Proof. See Appendix F. □

The mechanism is as follows. An increase in $m(a, a')$ raises the effective variety of inputs available to buyer a' , lowering its cost shifter $c(a')$ through love of variety. Lower costs raise revenue $r(a')$, which increases firm value $V(a')$ and, through the R&D first-order condition, the innovation rate $\lambda(a')$.

By Assumption 1, when supplier a innovates, its new products become available to existing buyer a' , increasing $m(a, a')$. The proposition therefore implies that suppliers'

innovation raises buyers' innovation.

The IV estimates in Section 2.3 provide empirical support for this prediction. Using foreign patent shocks as instruments for trading partners' R&D, we find that an exogenous increase in suppliers' R&D intensity raises the focal firm's R&D intensity. This is precisely what Proposition 4 predicts: suppliers' innovation expands the varieties available to buyers, improving their efficiency and strengthening their incentives to innovate.

4.5 Decomposition Tool: Second-Best Constrained Planner

The Social Planner corrects both market power and the externality simultaneously. To decompose their contributions, we introduce the Constrained Planner: a second-best benchmark that internalizes the externality only, while maintaining markups.

The decomposition works as follows:

- SP vs DE = Total inefficiency
- CP vs DE = Contribution of the externality alone
- SP vs CP = Contribution of market power alone

This decomposition is central to interpreting our quantitative results in Section 5.

The Constrained Planner maximizes the same objective function (18) as the First-Best Social Planner:

$$U_0^{CP} = \int_0^\infty \exp(-\rho t) \log Y(t) dt \quad (23)$$

subject to maintaining the decentralized goods market structure from Section 3: (6), (8), (9), (13), (14), and (15). The Constrained Planner optimally allocates only R&D workers $l_H(a)$ to address innovation externalities while preserving the static goods market distortions characterized by the decentralized markup structure.

Proposition 5 (Constrained Planner's R&D Allocation). *The Constrained Planner's allocation of R&D workers satisfies:*

$$l_H^{CP}(a) \propto [V^{CP}(a)]^{\frac{\gamma}{\gamma-1}} \quad (24)$$

where the constrained social value function $V^{CP}(a)$ solves:

$$\begin{aligned} \rho V^{CP}(a) = & r(a) - \delta_F V^{CP}(a) + V_a^{CP}(a) \\ & + [\lambda^{CP}(a) V^{CP}(a) - w_H^{CP} \phi(\lambda^{CP}(a))] \end{aligned}$$

$$+ \int V^{M,CP}(a, a') \left\{ \frac{\zeta}{N_f} f(a') - \delta_M m(a, a') \right\} da' \quad (25)$$

where $r(a) = \left(\frac{\mu c(a)}{P} \right)^{1-\sigma}$ $D(a)$ uses the decentralized demand and cost shifters $D(a)$ and $c(a)$ from equations (8) and (9), respectively, and w_H^{CP} denotes the corresponding shadow wage on R&D labor.

The Constrained Planner represents a realistic policy scenario where governments can influence R&D allocation through subsidies or taxes, but cannot directly intervene in goods market pricing decisions due to informational or institutional constraints.

This completes the theoretical framework. We have identified two sources of R&D misallocation and constructed tools to decompose their contributions. The key question remains: how large are these effects quantitatively? We turn to this in the next section.

5 Quantitative Analysis

This section presents the estimation of our supply chain innovation model with network-formation externalities using Japanese firm-level data, providing empirical validation of the theoretical framework developed in Section 4. We employ a two-stage approach: first, we estimate six structural parameters using Simulated Method of Moments (SMM) under the decentralized equilibrium characterized by Proposition 2, targeting moments from the observed age-based matching matrix. Second, we use these estimated parameters as inputs to solve for counterfactual allocations under the alternative institutional arrangements analyzed in Section 4, specifically the First-Best Social Planner and Second-Best Constrained Planner solutions from Propositions 3 and 5.

This methodology allows us to quantify the welfare implications of the two channels of R&D misallocation identified in Section 4: distortions from market power and network formation externalities. By comparing the estimated decentralized equilibrium with the theoretical benchmarks, we can isolate the welfare effects from addressing different types of market failures while maintaining structural consistency across all policy environments.

5.1 Parameter Estimation

5.1.1 Estimation Strategy

Our estimation strategy follows a two-stage approach designed to isolate the welfare effects of policy interventions. In the first stage, we estimate structural parameters under the decentralized equilibrium using Simulated Method of Moments (SMM), targeting moments

from the age-based matching matrix that characterizes network formation patterns in Japanese supply chains. The model is estimated using 100 Halton draws to reduce simulation variance, with the objective function minimizing the weighted distance between empirical and model-implied moments.

The parameter vector to be estimated is $\theta = (\zeta, \zeta_0, \delta_M, \gamma, \phi, \delta_F)$, where ζ represents the network formation efficiency parameter, ζ_0 captures the initial network density at firm entry, δ_M is the network link destruction rate, γ determines the R&D cost curvature, ϕ measures R&D productivity, and δ_F represents the firm exit rate.

In the second stage, we take the estimated parameter vector $\hat{\theta}$ as given and solve for equilibrium allocations under the alternative institutional arrangements analyzed in Section 4. This approach ensures that all welfare comparisons are conducted under identical structural parameters, isolating the pure effect of policy interventions. The First-Best Social Planner solution from equation (19) represents the first-best allocation that eliminates market power distortions and internalizes the network-formation externality, while the Second-Best Constrained Planner from equation (24) addresses only the network-formation externality while preserving the distortions from market power present in the decentralized market structure.

5.1.2 Estimated Parameters

The following parameters were estimated using Simulated Method of Moments targeting moments from the age-based matching matrix that characterizes network formation patterns in Japanese supply chains.

The network formation rate $\zeta = 0.3165$ indicates moderate efficiency in supply chain matching, while the high initial connectivity $\zeta_0 = 0.7371$ suggests new firms enter with substantial existing networks.

The innovation parameters reveal substantial convexity in R&D costs ($\gamma = 2.13$) and low base productivity ($\phi = 0.0004$), consistent with high-risk, high-return industrial innovation. The labor share parameter $\beta = 0.33$ captures the double marginalization structure in which upstream suppliers receive one-third of value added while downstream firms retain two-thirds.

The externally calibrated parameters are set based on previous literature and institutional knowledge. The elasticity of substitution $\sigma = 3$ follows Miyauchi (2024), representing moderate demand substitutability between intermediate goods. The discount rate $\rho = 0.05$ reflects a standard value commonly used in the literature for firm-level time preferences. The entry mass $\lambda_E = 1$ serves as a normalization of the firm entry flow. Under monopolistic competition with constant elasticity of substitution, the implied markup is $\mu = \sigma/(\sigma - 1) =$

Table 3: Model Parameters

Parameter	Symbol	Value
<i>Internally Estimated Parameters</i>		
Network formation rate	ζ	0.3165
Initial connectivity	ζ_0	0.7371
Link destruction rate	δ_M	0.0100
R&D curvature	γ	2.1316
R&D efficiency	ϕ	0.0003842
Firm exit rate	δ_F	0.0400
<i>Externally Calibrated Parameters</i>		
Elasticity of substitution	σ	3
Labor share	β	0.33
Discount rate	ρ	0.05
Entry mass	λ_E	1

Notes: Internal parameters estimated using Simulated Method of Moments targeting 16 moments from the age-based matching matrix M_{aa} . SMM estimation uses 100 Halton draws to reduce simulation variance.

1.5, capturing the price–cost margin that emerges from optimal pricing behavior.

5.1.3 Data Construction and Age Ranking

Before presenting model fit results, we describe the construction of the empirical matching matrix used in estimation. Firms are classified into age quantiles rather than continuous age values to ensure sufficient observations in each cell and to facilitate comparison with model predictions.

The empirical matching matrix is constructed as follows: First, firms are ranked into age quantiles based on their founding year, creating discrete age categories that ensure balanced sample sizes across groups. Second, for each firm pair connected in the supply chain network, we record both buyer and seller connections and compute the average connection strength between age groups. Third, the resulting density matrix is normalized so that all entries sum to one, representing the probability distribution of connections across age pairs.

5.1.4 Model Fit

Figure 5 presents the model’s fit to the targeted moments through side-by-side 3D visualizations of the empirical and simulated matching matrices. The model successfully captures the age-based network formation patterns, with some systematic deviations for

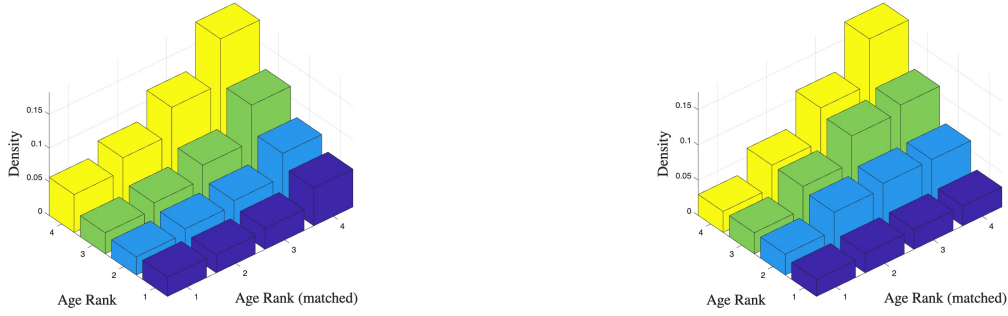


Figure 5: Model Fit: Empirical (left) vs Simulated (right) Matching Matrices

Notes: The 3D surfaces show the density of network connections between firms of different age categories.

very young and very old firms.

The model achieves a reasonable fit for most moments, capturing the essential features documented in Section 2 while exhibiting some systematic deviations. The empirical analysis reveals several patterns that the model successfully replicates: (1) the number of suppliers and buyers increases monotonically with firm age, but with decreasing rates of growth as firms mature; (2) firms exhibit positive age-assortative matching, where older firms preferentially connect with other older firms; and (3) business relationships display substantial stickiness, with more than 10 years required for 90% of relationships to dissolve.

The model successfully captures these age-dependent network formation patterns. The positive age-assortative matching observed in the data emerges naturally from the combination of random matching and relationship stickiness. As documented in Section 2, while new relationship formation shows little age dependence, the persistence of existing relationships causes firms and their partners to "age together," creating the observed assortativity.

The lifecycle patterns of innovation activity are also well-replicated. Section 2 documents that both patent counts and R&D/sales ratios are monotonically increasing functions of firm age, consistent with the model's prediction that innovation incentives grow with network connectivity. The empirical finding that connected firms' R&D activities positively affect their partners' innovation provides validation for the network-formation externality mechanisms central to the model.

However, some model limitations emerge in the precise quantitative fit. The random matching assumption may not fully capture the complex partner selection processes observed in reality, particularly for very young firms who may have access to specialized network formation channels. Additionally, the model's assumption of symmetric relationship formation may oversimplify the heterogeneous search and matching costs that vary

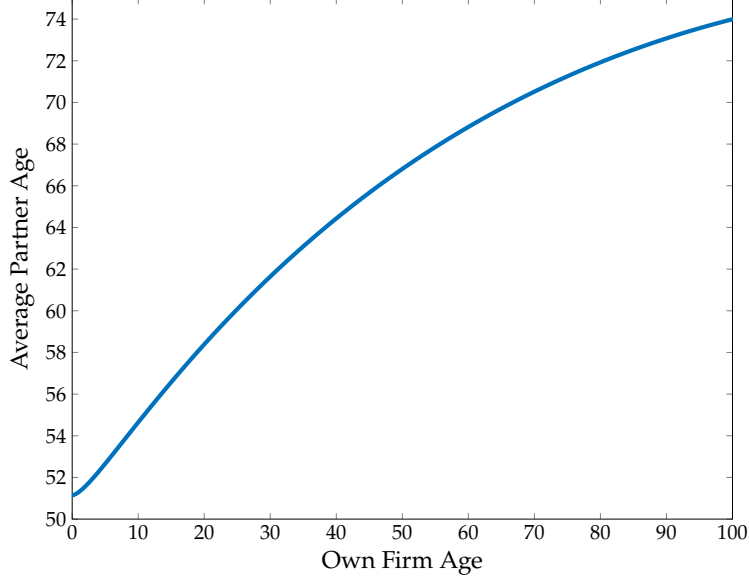


Figure 6: Age Assortativity in Business Networks

Notes: Average partner age plotted against a firm's own age. The 45-degree line represents perfect age assortativity.

across firm ages and industries.

To further verify that our model captures the empirical patterns documented in Section 2, we examine the targeted moments from Figure 5 from different perspectives. Figure 6 presents an alternative visualization of the age-based matching patterns, confirming that the model successfully replicates the positive age assortativity observed in the Section 2 data, where firms tend to form partnerships with other firms of similar ages. While the model exhibits slightly younger firm ages compared to the Section 2 data, it successfully captures the main empirical patterns, including the age assortativity structure and matching density distribution observed in Japanese supply chain networks.

Figure 7 provides another perspective on the matching patterns, examining how network connectivity and production efficiency evolve jointly over the firm lifecycle. The left panel shows the mass of connections and the right panel shows the growth rate of connections. This pattern confirms the finding from Section 2 that mature firms achieve higher productivity through extensive supplier relationships, with both the mass and growth rate of connections exhibiting the tendencies documented in the empirical analysis.

5.2 Lifecycle Dynamics and R&D Misallocation

Our quantitative model, consistent with the empirical evidence in Section 2, generates systematic patterns in how network positions and innovation incentives co-evolve over the

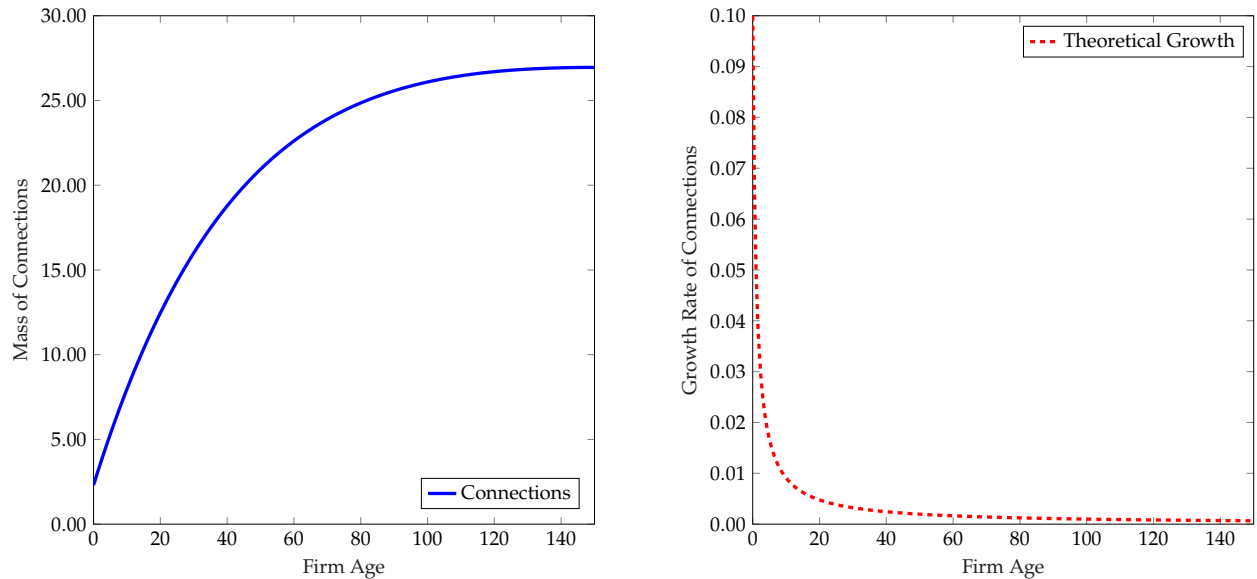


Figure 7: Lifecycle Production Network Analysis

Notes: Left panel shows mass of connections by firm age. Right panel shows growth rate of connections by firm age.

firm lifecycle. These dynamics are the key to understanding the origins and nature of R&D misallocation.

5.2.1 Network Formation and Innovation Over the Lifecycle

Beyond the matching moments targeted in estimation, the model generates lifecycle patterns for innovation and productivity that align with the data. Figure 8 illustrates that innovation rates rise with age as firms build larger and more valuable supplier networks. This creates a powerful complementarity between a firm's network position and its incentive to innovate. As firms age, the model's dynamic process of random matching and link stickiness leads them to accumulate a larger mass of connections, as documented in Figure 1a. The emergent network structure produces the positive age-assortative matching shown in Figure 6, where older firms disproportionately connect with other mature firms. This pattern drives two key mechanisms underlying higher productivity among older firms: first, the *love of variety effect* from accessing a broader range of intermediate inputs, and second, the *quality composition effect* from preferentially matching with more productive mature suppliers.

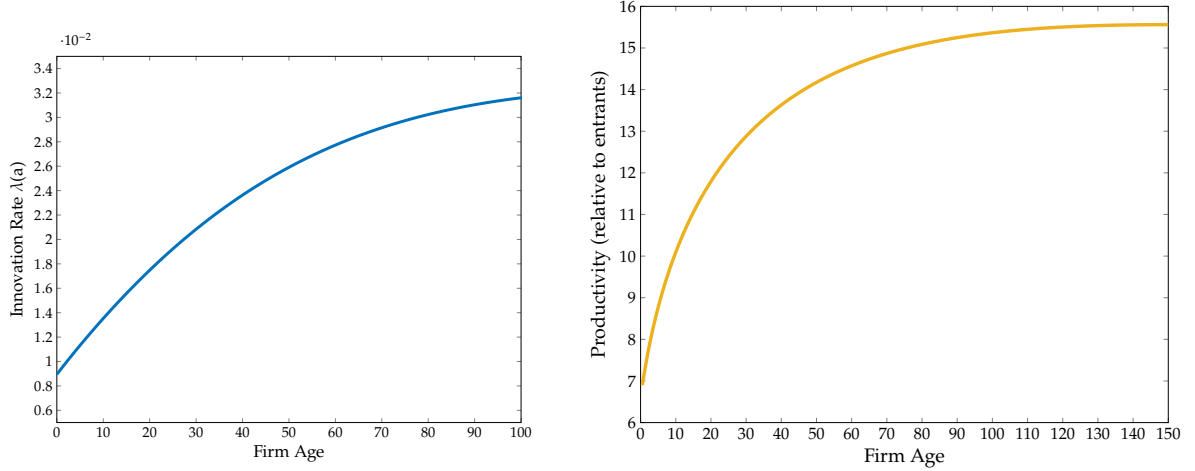


Figure 8: Firm Lifecycle Dynamics

Notes: Innovation rates increase with age (left panel), leading to productivity accumulation over the firm lifecycle (right panel). Productivity is defined as the inverse of marginal cost, $q(a) = 1/c(a)$, normalized relative to entrants. The patterns reflect the complementarity between network position and innovation incentives.

Table 4: Welfare Gains

Environment	Welfare Gain relative to DE
Social planner	+35.84%
Constrained planner	+0.09%

Notes: The table reports the consumption welfare gain relative to the decentralized equilibrium

5.2.2 Welfare Analysis and Sources of R&D Misallocation

Figure 9 reveals a striking pattern: older firms occupy central positions in production networks with more buyers and suppliers, generating higher sales levels, yet their revenues remain disproportionately suppressed relative to the socially optimal level due to markup distortions that compound through supply chains. This suppression becomes more pronounced with age, reflecting how the cumulative effect of double marginalization particularly penalizes firms embedded in longer supply chains.

To analyze the welfare implications of this structure, we compare three policy environments as defined in Section 4: the *decentralized equilibrium* (DE), the *social planner* (SP), and the *constrained planner* (CP).

Table 4 presents the welfare outcomes, while Figures 10a and 10b show the corresponding R&D allocations. A key observation from Figure 10b is that the SP's optimal R&D allocation is remarkably similar to that of the DE. This finding allows us to infer the primary source of the large welfare gain reported in Table 4. Since the R&D allocation

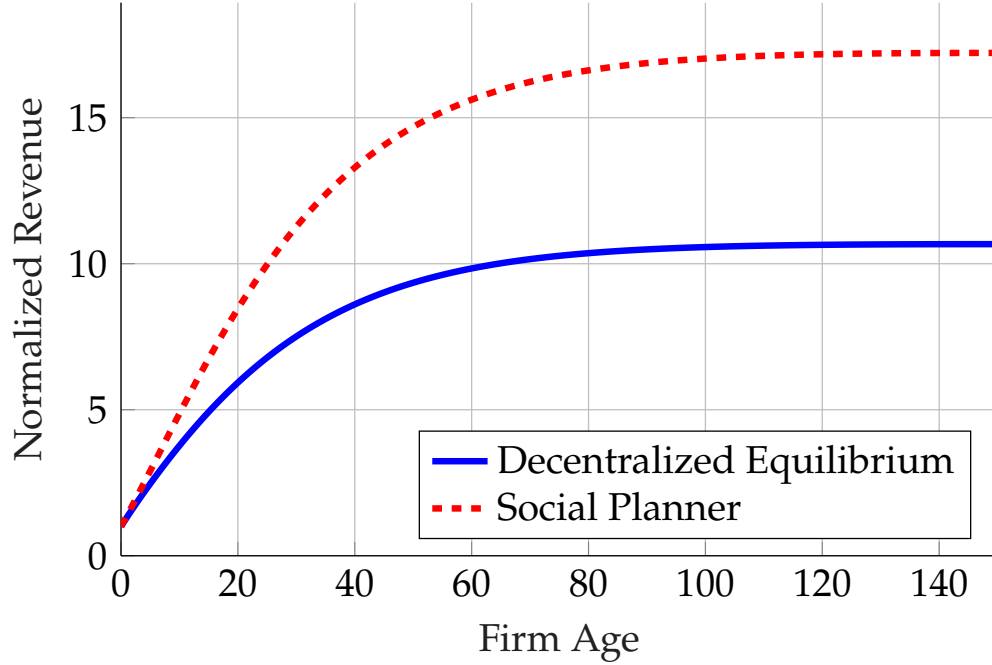


Figure 9: Revenue Comparison by Firm Age

Notes: Revenue patterns across firm age under Decentralized Equilibrium (DE) and Social Planner (SP) allocations. Both series are normalized to 1 at age 0.

barely changes, the substantial 35.84% gain under the SP must predominantly stem from correcting the *misallocation of production inputs* (general labor) that arises from eliminating markup distortions. The minimal welfare gain of 0.09% under the CP, where only R&D resources are reallocated, corroborates this conclusion.

This raises the central question for our analysis: if significant network-formation externalities are present, why does the SP's optimal R&D allocation so closely track the distorted decentralized outcome, especially when the CP allocation deviates significantly? The answer lies in the interaction of two powerful, counteracting forces that shape the SP's allocation.

The first force arises from the *correction of market power distortions*. In the DE, double marginalization distorts the allocation of both production labor and R&D labor by disproportionately suppressing the revenue and private value of innovation, $V(a)$, for older firms embedded in long supply chains. The SP, by eliminating markups as part of the transition from Proposition 2 to Proposition 3, boosts the relative value of these older firms. This effect, in isolation, creates a strong incentive to reallocate R&D resources *towards* older firms.

The second force is the *internalization of the network-formation externality*, which is a positive externality. This effect is isolated by the CP, whose optimal allocation is governed

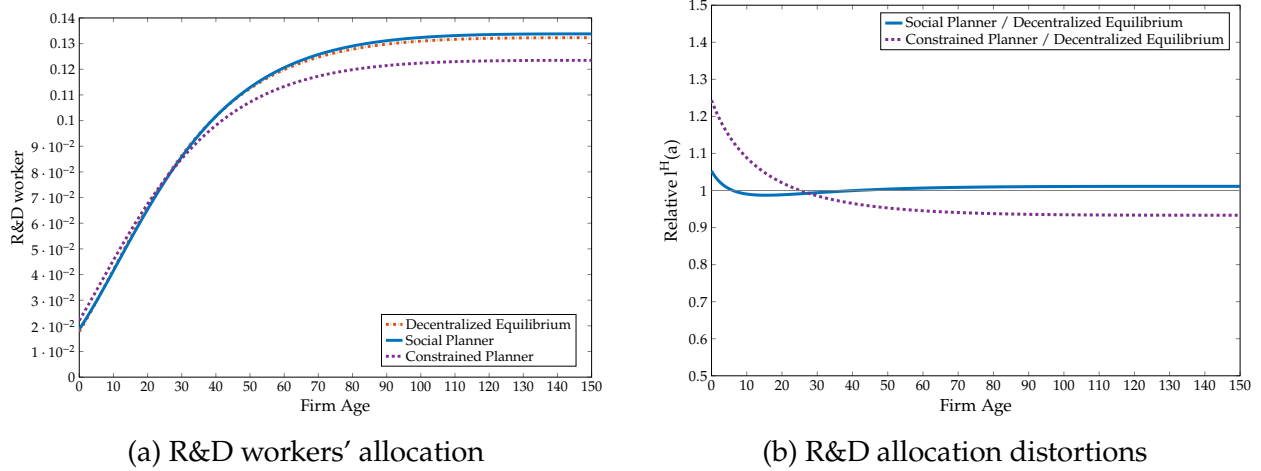


Figure 10: R&D allocation outcomes across policy environments

Notes: The left panel shows normalized R&D labor demand across the three policy environments. The right panel shows the relative allocation ratios to the decentralized equilibrium.

by Proposition 5. This allocation differs from the DE only by the inclusion of the network-formation externality wedge from equation (22):

$$\text{wedge}(a) = \int_0^\infty V^M(a, a') \left[\underbrace{\frac{\zeta f(a')}{N_f}}_{\text{Formation}} - \underbrace{\delta_M m(a, a')}_{\text{Destruction}} \right] da'$$

Here, $V^M(a, a')$ is the total social value generated by the trading relationship between a supplier of age a and a buyer of age a' . The flow value of the match, $(1 - \beta) \cdot r^{SP}(a') \cdot \Omega(a, a')$, represents the net love of variety benefit after accounting for resource costs. In the primitive form of the Social Planner's problem, this value derives from two terms: gross love of variety benefit for buyer a' minus production resource cost for supplier a . Under CES optimality, these combine into the single expression shown, where $\Omega(a, a')$ is supplier a 's expenditure share in buyer a' 's intermediate input bundle. The match value is determined by its own Hamilton-Jacobi-Bellman equation from Proposition 3:

$$(\rho + \delta_F + \delta_M - \lambda(a)) V^M(a, a') = V_a^M(a, a') + V_{a'}^M(a, a') + (1 - \beta) \cdot r^{SP}(a') \cdot \Omega(a, a')$$

where $\Omega(a, a') = \frac{c(a)^{1-\sigma}}{\int c(a'')^{1-\sigma} m(a'', a') da''}$ is the expenditure share of supplier a in buyer a' 's total intermediate consumption.

The wedge thus quantifies the value a firm of age a fails to internalize at the two margins

of relationship dynamics. The first term inside the integral is the formation component, representing the gross inflow of new buyer relationships. When a match occurs, buyer a' gains love of variety benefits from accessing supplier a 's products, creating social surplus $V^M(a, a')$, but the innovating firm a neglects the portion accruing to its new buyer a' . The second term is the destruction component, representing the gross outflow of dissolving relationships. When a link is lost, the love of variety benefits $V^M(a, a')$ are destroyed, but firm a externalizes the loss incurred by its buyer.

As shown by the dotted line in Figure 10b, internalizing this wedge alone, as the CP does, reallocates R&D resources *away* from the oldest firms relative to the DE. This direction results from the magnitude of the network-formation externality depending on the *net flow* of match formation. Young firms are in a phase of rapid network expansion: the rate at which they form new buyer relationships exceeds the rate of relationship destruction. Each new match creates love of variety benefits for the buyer that the young supplier does not internalize. Old firms, by contrast, have mature networks where formation and destruction roughly balance. Their marginal R&D generates limited additional love of variety spillovers. Because this is a positive externality, the underinvestment in R&D is most severe for young firms.

Specifically, the wedge decomposes as:

$$\text{wedge}(a) = \int V^M(a, a') \cdot \underbrace{\left[\frac{\zeta}{N_f} f(a') - \delta_M m(a, a') \right]}_{\text{Net formation flow}} da'$$

For young suppliers (low a), existing match stock $m(a, a')$ is small, so the destruction term is small and net formation is large. For old suppliers (high a), $m(a, a')$ is large, so destruction nearly cancels formation. This explains why the Constrained Planner reallocates R&D toward younger firms.

In the Social Planner's solution, these two forces act simultaneously and in opposite directions on older firms. The push to allocate *more* resources to them to correct for market power distortions is largely offset by the push to allocate *less* to them due to the specifics of the network-formation externality. This cancellation explains why the SP's final R&D allocation appears close to the original decentralized one. The CP's environment, lacking the first force, reveals the unmitigated impact of the second, explaining its significant deviation from the DE.

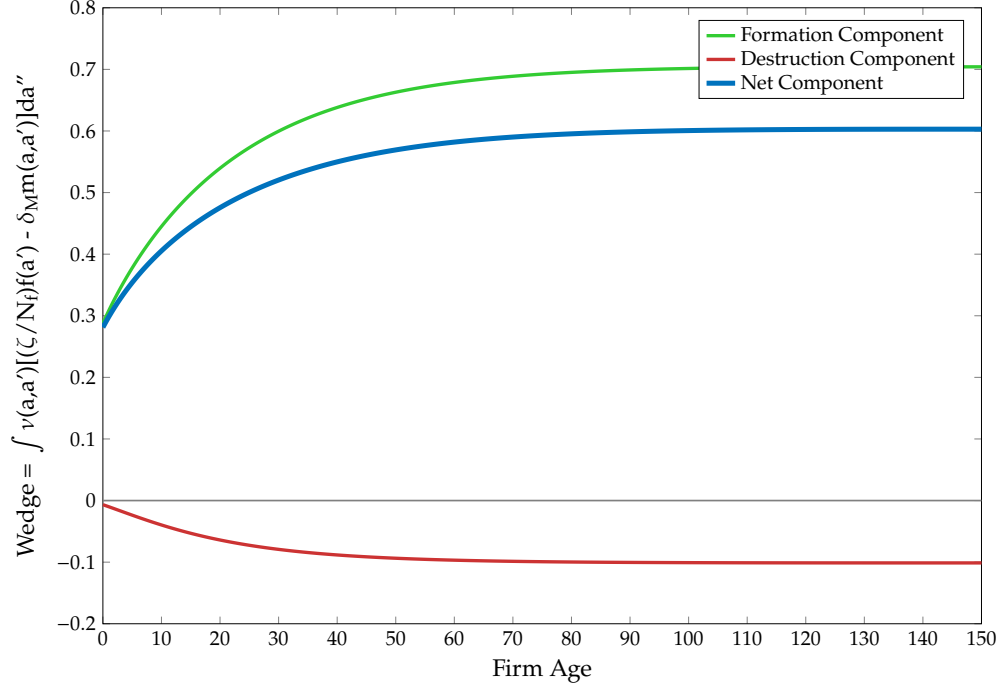


Figure 11: Network-Formation Externality Wedge Decomposition

Notes: Decomposition of the network-formation externality wedge. The formation component represents $\int V^M(a, a') \frac{\zeta f(a')}{N_f} da'$: the love of variety benefits created when supplier a forms new matches with buyers. The destruction component represents $\int V^M(a, a') \delta_{MM}(a, a') da'$: the love of variety benefits lost when existing matches dissolve. The net component shows that young suppliers create larger externalities because they are in a phase of rapid network expansion.

5.2.3 Policy Implications

Our quantitative results offer a clear policy implication, though one with important trade-offs. The minimal welfare gain under the Constrained Planner (0.09%) demonstrates that correcting R&D misallocation in isolation yields negligible benefits. This suggests that the optimal policy focus should be on addressing the large welfare losses stemming from the misallocation of production inputs due to market power.

However, a policy focused solely on eliminating market power could have a counterintuitive side effect on the allocation of R&D. As the Constrained Planner's solution reveals, correcting only for the network-formation externality pushes the R&D allocation for older firms even further from the decentralized equilibrium than the first-best allocation does (Figure 10b). This implies that the optimal policy of removing markups might be accompanied by an amplification of the R&D misallocation. Nevertheless, our findings suggest this is a worthwhile trade-off. The quantitative impact of the R&D resource misallocation on aggregate welfare is vanishingly small compared to the large, first-order

gains from correcting the misallocation of production inputs.

6 Conclusion

This paper investigates how production networks shape firms' R&D decisions, identifying and quantifying the aggregate inefficiencies that arise from this relationship. Guided by empirical patterns from Japanese firm-level data, we build a dynamic model where firms establish supplier networks through an exogenous matching process and leverage these connections to introduce new products.

Our framework uncovers a novel positive externality, the *network-formation externality*, arising from love of variety: when firms innovate, they become more attractive trading partners with richer product lines, yet the value of this improvement goes uncompensated. Because this is a positive externality, private R&D falls short of the social optimum. When we take this model to the data, our central finding is that this externality interacts with market power distortions in a crucial way. The optimal R&D allocation in a first-best world is nearly identical to the decentralized allocation. This is because two counteracting forces on older firms almost perfectly offset each other: correcting for markups pushes R&D resources towards them, while internalizing the positive externality pushes resources away.

Echoing the policy discussion above, this result points to a clear takeaway. Because the forces governing R&D allocation cancel out, the optimal policy derives its substantial welfare gains not from reallocating R&D, but from correcting the misallocation of *production inputs* due to market power. Our findings underscore the importance of understanding the general equilibrium interactions between market failures. Policies that target dynamic inefficiencies like R&D misallocation in isolation, without addressing the large, underlying distortions from market power, may have limited impact on aggregate welfare.

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Appendix for “Production Networks and R&D Allocation”

A Additional Empirical Results

A.1 Cross-sectional correlation between production and citation networks

In this section, we show that the production and citation networks overlap at the firm level. Using the combined TDB and IIP patent databases, we find that citation relationships in Japan are strongly associated with production links. Conditional on an observed transactional relationship, the probability of a citation tie is roughly 30 times larger than under a random benchmark. Moreover, this probability declines monotonically with network distance, underscoring the importance of distinguishing the two networks when working with firm-level data.

The integrated database covers Japanese firms that filed at least one patent application between 1998 and 2019. We construct a patent-citation matrix with citing firms in rows and cited firms in columns, where an entry of one indicates the presence of a citation link. Analogously, we build an input-output matrix that records inter-firm production relationships among patenting firms; the columns index suppliers, the rows index purchasers, and a value of one denotes an active business relationship. These matrices enumerate the potential links between Japanese patenting firms, which we use to study how production and patent networks interact.

Panel (a) summarizes the estimates from a linear probability model relating direct production links to patent citations. Columns (1)–(4) vary the set of control variables. The estimated coefficient implies that, conditional on a direct business relationship, the probability of observing a citation link rises by about 35% relative to the random benchmark.

The right column of panel (a) examines higher-order production-network linkages. The probability of a citation link declines steadily with network distance: second- and third-order links remain relevant, whereas fourth-order and more distant connections are nearly irrelevant. Panel (b) provides a visual summary of this decay with respect to production-network distance.

Table A1: Summary of Production and Patent Network Links

Number of Patent Firms	10,481	
Possible Combinations of Firms	109,851,361	
Active Patent Filing Links	2,548,881	(2.3%)
Active Production Network Links	1,647,770	(1.5%)

Alternative measurement

To address concerns about the parametric assumptions of the linear probability model, we also compute conditional probabilities for different subsamples $g \in G$,

$$P_C \equiv \Pr(\text{patent connection} = 1 \mid \text{network connection} = 1, g).$$

We estimate them empirically as

$$\hat{P}_C = \frac{|\text{patent connection} = 1 \cap \text{network connection} = 1, g|}{|\text{network connection} = 1, g|},$$

and compare them with the corresponding unconditional probabilities,

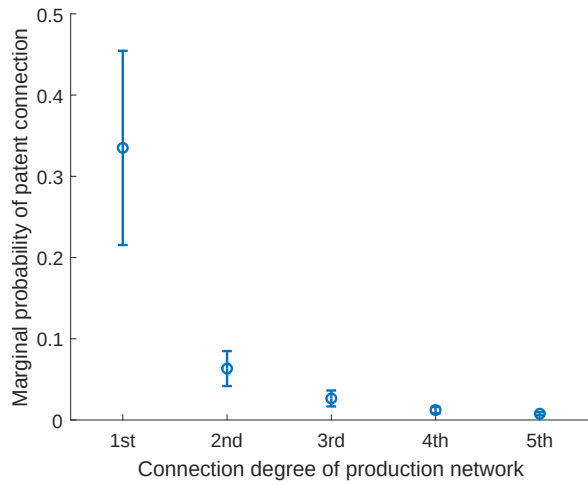
$$\hat{P}_U = \frac{|\text{patent connection} = 1, g|}{|g|}.$$

We form subsamples using every combination of firm-pair characteristics: industry, firm-size decile, and location. Figure A1 plots the conditional probabilities against the corresponding unconditional probabilities for each group.

Across all groupings, the unconditional probability is close to zero because the matrices are very sparse, whereas the average conditional probability remains around 30–40% given a production-network connection.

Table A2: Production Network and Patent Connection

(a) Network degrees and patent connection



Notes:

	First Degree Connection				Higher Order Degree Connection			
	(1)	(2)	(3)	(4)	Second	Third	Fourth	Fifth
network connection	0.343*** (0.062)	0.339*** (0.062)	0.344*** (0.062)	0.335*** (0.061)	0.063*** (0.011)	0.026*** (0.005)	0.012*** (0.002)	0.007*** (0.001)
Industry Pair FE	✓			✓	✓	✓	✓	✓
Size Pair FE		✓		✓	✓	✓	✓	✓
Prefecture Pair FE			✓	✓	✓	✓	✓	✓
Observations	93161103	93161104	93161104	93161103	93161103	93161103	93161103	93161103

Standard errors in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Panel (b) reports linear probability estimates that relate the possible combinations of patent firm-to-firm networks in Table A1 to the presence of production-network links, controlling for firm-pair characteristics: (a) two-digit Japanese Standard Industrial Classification, (b) firm-size deciles, and (c) prefectures. Panel (a) plots the corresponding coefficients.

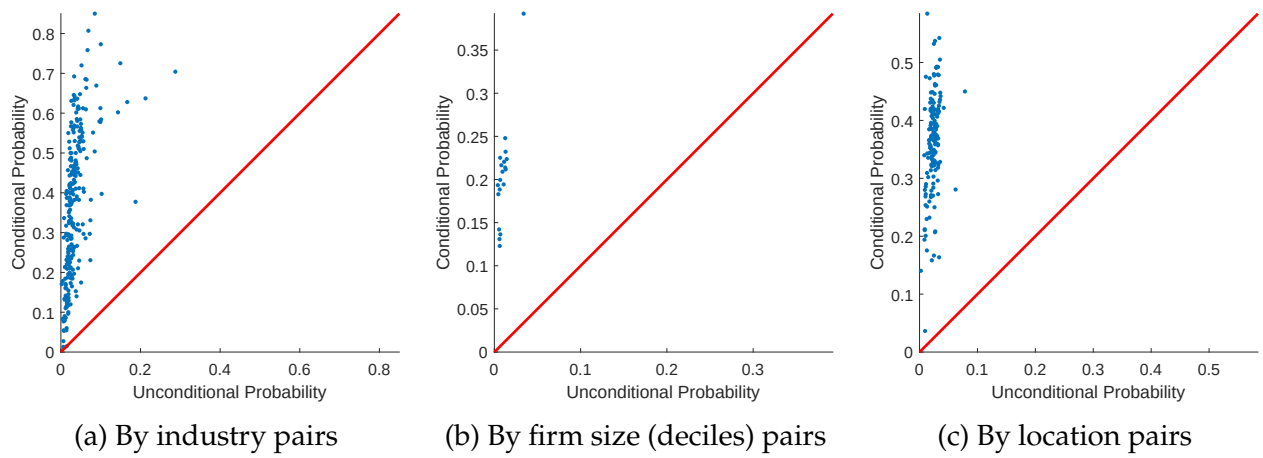


Figure A1: Conditional and unconditional probability of patent connection.

Notes: The x-axis reports the unconditional probability $\frac{|\text{patent connection}=1, g|}{|g|}$, and the y-axis shows the conditional probability $\frac{|\text{patent connection}=1 \cap \text{network connection}=1, g|}{|\text{network connection}=1, g|}$. The groups $g \in G$ include all combinations of firm-pair characteristics: (a) two-digit Japan Standard Industrial Classification, (b) firm-size deciles, and (c) prefectures. Groups with fewer than 50 observations in either patent relationships or production-network connections are excluded.

B Equation list of Equilibrium

The list of equations for solving the steady state equilibrium is as follows: (i) the value of product $V(a)$; (ii) the innovation rates $\lambda(a)$; (iii) the distribution of products $f(a)$; (iv) the distribution of matched buyers and suppliers $m(a', a)$; (v) the cost function $c(a)$; (vi) the demand shifter $D(a)$; (vii) the wage for production worker w ; (viii) the wage for R&D worker w_H , and (ix) the price index P

1. The value of products:

$$(\rho + \delta_F - \lambda(a)) V(a) = \left(1 - \frac{1}{\mu}\right) r(a) + V_a(a) - w_H \phi(\lambda(a))$$

where revenue, $r(a)$ is

$$r(a) = \left(\frac{\mu c(a)}{P}\right)^{1-\sigma} D(a)$$

and the price index, P is

$$P = \mu \left(\int c(a)^{1-\sigma} dF(a) \right)^{\frac{1}{1-\sigma}}$$

2. FOC for the innovation rate:

$$\lambda(a) = \left\{ \frac{\phi}{\gamma' w_H} V(a) \right\}^{\frac{1}{\gamma-1}} \quad (26)$$

3. The distribution of products $f(a)$:

$$0 = -\frac{\partial}{\partial a} f(a) + (\lambda(a) - \delta_F) f(a) + \lambda_E \delta(a) \quad (27)$$

4. The distributions of matched products $m(a', a)$:

$$\begin{aligned} \frac{\partial}{\partial a} m(a', a) = & -\frac{\partial}{\partial a'} m(a', a) - (\delta_M + \delta_F) m(a', a) + \frac{\zeta}{N_f} f(a') \\ & + \zeta_0 \frac{\lambda_E}{N_f} \delta(a) + \lambda(a') m(a', a) \end{aligned} \quad (28)$$

subject to the boundary condition $m(a'; 0) = \zeta_0 f(a')$.

5. The cost shifter $c(a)$ satisfies

$$c(a) = w^\beta \left(\int (\mu c(a'))^{1-\sigma} m(a', a) da' \right)^{\frac{1-\beta}{1-\sigma}} \quad (29)$$

where $w = \frac{\beta}{\mu - (1-\beta)}$.

6. The demand shifter for an age a product $D(a)$ satisfies

$$D(a) = (1 - \beta) \mu^{-\sigma} \int \left[\frac{c(a')}{w} \right]^{\frac{\beta}{1-\beta}(\sigma-1)} D(a') m(a', a) da' + 1 \quad (30)$$

7. The skilled wage w_H satisfies the skilled labor market clearing condition

$$1 = \int \phi(\lambda(a)) f(a) da \quad (31)$$

Equation list of social planner solution:

1. Social value function

$$\begin{aligned} \left(\rho + \delta_F - \lambda^{SP}(a) \right) V^{SP}(a) = & r^{SP}(a) + \frac{\partial}{\partial a} V^{SP}(a) - w_H^{SP} \phi(\lambda^{SP}(a)) \\ & + \int V^M(a, a') \left\{ \frac{\zeta}{N_f} f(a') - \delta_M m(a, a') \right\} da', \end{aligned}$$

where

$$r^{SP}(a) = \left(\frac{c^{SP}(a)}{P^{SP}} \right)^{1-\sigma} D^{SP}(a)$$

$$P^{SP} = \left(\int c^{SP}(a)^{1-\sigma} f(a) da \right)^{\frac{1}{1-\sigma}}$$

$$\begin{aligned} \left(\rho + \delta_F + \delta_M - \lambda^{SP}(a) \right) V^M(a, a') = & \frac{\partial}{\partial a} V^M(a, a') + \frac{\partial}{\partial a'} V^M(a, a') \\ & + r^{SP}(a') (1 - \beta) \Omega(a, a') f(a') \end{aligned}$$

$$\Omega(a, a') = \frac{c^{SP}(a)^{1-\sigma}}{\int c^{SP}(a'')^{1-\sigma} m(a'', a') da''}$$

2. FOC for the innovation rate:

$$\lambda^{SP}(a) = \left\{ \frac{\phi}{\gamma w_H^{SP}} V^{SP}(a) \right\}^{\frac{1}{\gamma-1}} \quad (32)$$

3. The distribution of products $f(a)$:

$$0 = -\frac{\partial}{\partial a} f(a) + \left(\lambda^{SP}(a) - \delta_F \right) f(a) + \lambda_E \delta(a) \quad (33)$$

4. The distribution of matched products $m(a', a)$:

$$\begin{aligned} \frac{\partial}{\partial a} m(a', a) = & -\frac{\partial}{\partial a'} m(a', a) - (\delta_F + \delta_M) m(a', a) + \frac{\zeta}{N_f} f(a') \\ & + \zeta_0 \frac{\lambda_E}{N_f} \delta(a) + \lambda^{SP}(a') m(a', a) \end{aligned} \quad (34)$$

subject to the boundary condition $m(a'; 0) = \frac{\zeta_0}{N_f} f(a')$.

5. The cost shifter $c(a)$ satisfies

$$c^{SP}(a) = \left(\int \left(c^{SP}(a') \right)^{1-\sigma} m(a', a) da' \right)^{\frac{1-\beta}{1-\sigma}} \quad (35)$$

6. The demand shifter for an age a product $D(a)$ satisfies

$$D^{SP}(a) = (1 - \beta) \int c(a')^{SP \frac{\beta}{1-\beta} (\sigma-1)} D^{SP}(a') m(a', a) da' + 1 \quad (36)$$

7. The skilled wage w_H^{SP} satisfies the skilled labor market clearing condition

$$1 = \int \phi \left(\lambda^{SP}(a) \right) f(a) da \quad (37)$$

C Proofs and derivation

Proof of Lemma 1

Given the production technology of intermediate good production, firm ω has the following marginal cost:

$$c(\omega) = w^\beta \left(\int_{\omega' \in \mathcal{S}(\omega)} p(\omega', \omega)^{1-\sigma} d\omega' \right)^{\frac{1-\beta}{1-\sigma}}$$

Inserting $p(\omega', \omega) = \mu c(\omega')$ and rewriting the integral using the matched product density $m(a', a)$, we get

$$c(a) = w^\beta \left(\int (\mu c(a'))^{1-\sigma} m(a', a) da' \right)^{\frac{1-\beta}{1-\sigma}}$$

Using Shephard's Lemma, Hicksian demand for ω from ω' can be obtained by

$$\begin{aligned} x(\omega, \omega') &= \frac{\partial c(\omega') x(\omega')}{\partial p(\omega)} \\ &= \frac{\partial \log c(\omega') x(\omega')}{\partial p(\omega)} c(\omega') x(\omega') \\ &= \frac{\partial}{\partial p(\omega)} \left[\log \left\{ w^\beta \left(\int_{\omega'' \in \mathcal{S}(\omega')} p(\omega'')^{1-\sigma} d\omega'' \right)^{\frac{1-\beta}{1-\sigma}} x(\omega') \right\} \right] c(\omega') x(\omega') \\ &= \frac{1-\beta}{1-\sigma} \frac{\partial}{\partial p(\omega)} \left[\log \left(\int_{\omega'' \in \mathcal{S}(\omega')} p(\omega'')^{1-\sigma} d\omega'' \right) \right] c(\omega') x(\omega') \\ &= \frac{1-\beta}{1-\sigma} \frac{(1-\sigma) p(\omega)^{-\sigma}}{\int_{\omega'' \in \mathcal{S}(\omega')} p(\omega'')^{1-\sigma} d\omega''} c(\omega') x(\omega') \\ &= (1-\beta) \frac{p(\omega)^{-\sigma}}{\int_{\omega'' \in \mathcal{S}(\omega')} p(\omega'')^{1-\sigma} d\omega''} c(\omega') x(\omega') \end{aligned} \tag{38}$$

Note that

$$\begin{aligned} c(\omega) &= w^\beta \left(\int_{\omega' \in \mathcal{S}(\omega)} p(\omega')^{1-\sigma} d\omega' \right)^{\frac{1-\beta}{1-\sigma}} \\ c(\omega)^{\frac{1-\sigma}{1-\beta}} &= w^{\frac{\beta}{1-\beta}(1-\sigma)} \int_{\omega' \in \mathcal{S}(\omega)} p(\omega')^{1-\sigma} d\omega' \\ \int_{\omega' \in \mathcal{S}(\omega)} p(\omega')^{1-\sigma} d\omega' &= w^{\frac{\beta}{1-\beta}(\sigma-1)} c(\omega)^{\frac{1-\sigma}{1-\beta}} \end{aligned}$$

Inserting this into (38),

$$\begin{aligned} x(\omega, \omega') &= (1 - \beta) p(\omega)^{-\sigma} c(\omega') x(\omega') w^{-\frac{\beta}{1-\beta}(\sigma-1)} c(\omega')^{\frac{\sigma-1}{1-\beta}} \\ x(\omega, \omega') &= (1 - \beta) p(\omega)^{-\sigma} w^{-\frac{\beta}{1-\beta}(\sigma-1)} c(\omega')^{\frac{\sigma-\beta}{1-\beta}} x(\omega') \end{aligned}$$

From the good market clearing condition,

$$\begin{aligned} x(\omega) &= \int_{\omega' \in \mathcal{B}(\omega)} x(\omega, \omega') d\omega' + y(\omega) \\ &= \int_{\omega' \in \mathcal{B}(\omega)} (1 - \beta) p(\omega)^{-\sigma} w^{-\frac{\beta}{1-\beta}(\sigma-1)} c(\omega')^{\frac{\sigma-\beta}{1-\beta}} x(\omega') d\omega' + y(\omega) \\ &= (1 - \beta) p(\omega)^{-\sigma} w^{-\frac{\beta}{1-\beta}(\sigma-1)} \int_{\omega' \in \mathcal{B}(\omega)} c(\omega')^{\frac{\sigma-\beta}{1-\beta}} x(\omega') d\omega' + Y p(\omega)^{-\sigma} P^\sigma \end{aligned}$$

Using the matched product distribution function

$$x(a) = (1 - \beta) p(a)^{-\sigma} w^{-\frac{\beta}{1-\beta}(\sigma-1)} \int c(a')^{\frac{\sigma-\beta}{1-\beta}} x(a') m(a', a) da' + Y p(a)^{-\sigma} P^\sigma$$

Denote $D(a) = x(a) P^{1-\sigma} p(a)^\sigma$ as demand shifter for a ,

$$D(a) = (1 - \beta) \mu^{-\sigma} \int \left[\frac{c(a')}{w} \right]^{\frac{\beta}{1-\beta}(\sigma-1)} D(a') m(a', a) da' + PY$$

Finally, the ideal price index P is given by

$$\begin{aligned} P &= \left(\int_{\omega \in \Omega} p(\omega)^{1-\sigma} d\omega \right)^{\frac{1}{1-\sigma}} \\ P &= \left(\int (\mu c(a))^{1-\sigma} dF(a) \right)^{\frac{1}{1-\sigma}} \end{aligned}$$

Labor Demand and Labor Market Clearing for Production Worker

Using Shephard's Lemma, labor demand from ω can be obtained by

$$\begin{aligned} l(\omega) &= \frac{\partial c(\omega)}{\partial w} x(\omega) \\ &= \frac{\partial \log c(\omega) x(\omega)}{\partial w} c(\omega) x(\omega) \end{aligned}$$

$$\begin{aligned}
&= \frac{\partial}{\partial w} \left[\log \left\{ w^\beta \left(\int_{\omega' \in \mathcal{S}(\omega)} p(\omega')^{1-\sigma} d\omega' \right)^{\frac{1-\beta}{1-\sigma}} x(\omega) \right\} \right] c(\omega) x(\omega) \\
&= \beta \frac{\partial}{\partial w} [\log w] c(\omega) x(\omega) \\
&= \frac{\beta c(\omega) x(\omega)}{w}
\end{aligned}$$

With a notation,

$$\begin{aligned}
wl(a) &= \beta c(a) x(a) \\
&= \beta c(a) D(a) P^{\sigma-1} p(a)^{-\sigma} \\
&= \beta \mu^{-\sigma} \left(\frac{c(a)}{P} \right)^{1-\sigma} D(a)
\end{aligned}$$

From unskilled labor market clearing condition,

$$w = \beta \mu^{-\sigma} \int \left(\frac{c(a)}{P} \right)^{1-\sigma} D(a) dF(a)$$

Decomposition of Value Function

Conjecture that the value function takes an additive form

$$V^F(n, a) = nV(a).$$

Then, 10 can be expressed by

$$r n V(a) - n V_t(a) = n \pi(a) - \delta_F n V(a) + n V_a(a) + \max_{\lambda \geq 0} [n \lambda V(a) - n w_H \phi(\lambda)]$$

Dividing both side by n ,

$$r(t) V(a) - V_t(a) = \pi(a) - \delta_F V(a) + V_a(a) + \max_{\lambda \geq 0} [\lambda V(a, t) - w_H \phi(\lambda)]$$

Proof of Proposition 3 (i): Static Conditions

Proof. Let define $\mu_p(a) = Y^{\frac{1}{\sigma}} c^{SP}(a)$ and $\mu_l = Y^{\frac{1}{\sigma}}$ as Lagrangian multipliers for (??) and (??).

Then, first order conditions are

$$y(a) = c^{SP}(a)^{-\sigma}, \quad (39)$$

$$l(a) = \beta c(a)^{SP} x(a), \quad (40)$$

and

$$x(a', a) = (1 - \beta) [c(a)^{SP}]^{-\sigma} [c(a')^{SP}]^{\frac{\sigma-\beta}{1-\beta}} x(a') \quad (41)$$

By substituting (40) and (41) into (??), we have

$$x(a) = \frac{1}{\beta^\beta (1 - \beta)^{1-\beta}} [\beta c^{SP}(a) x(a)]^\beta \left(\int [(1 - \beta) c^{SP}(a)^{-\sigma} c^{SP}(a')^{\frac{\sigma-\beta}{1-\beta}} x(a')]^{\frac{\sigma-1}{\sigma}} m(a', a) da' \right)^{\frac{\sigma}{\sigma-1} (1-\beta)}.$$

Cancelling $x(a)$ from both sides and rearranging yields

$$c^{SP}(a) = \left(\int [c^{SP}(a')]^{1-\sigma} m(a', a) da' \right)^{\frac{1-\beta}{1-\sigma}},$$

Next, by substituting (40) and (41) into (??), and with similar step in the proof of Lemma 1, we could derive:

$$D^{SP}(a) = (1 - \beta) \int [c^{SP}(a')]^{\frac{\beta}{1-\beta}(\sigma-1)} D^{SP}(a') m(a', a) da' + 1,$$

where $D^{SP}(a) = x(a) [P^{SP}]^{1-\sigma} [c^{SP}(a)]^\sigma$. Finally, from (??) we have:

$$1 = \beta \int \left(\frac{c(a)^{SP}}{P^{SP}} \right)^{1-\sigma} D^{SP}(a) f(a) da$$

□

D Social Planner's Problem

We derive the Social Planner's optimal R&D allocation using the primitive form with $\log Y$ as the objective. This approach makes the economic interpretation of the network-formation externality transparent: the flow value of each match decomposes into the love of variety benefit for the buyer minus the resource cost for the supplier.

Setup

The Social Planner maximizes:

$$\max \int_0^\infty e^{-\rho t} \log Y(t) dt \quad (42)$$

where

$$Y = \left(\int y(a)^{\frac{\sigma-1}{\sigma}} f(a) da \right)^{\frac{\sigma}{\sigma-1}} \quad (43)$$

Production technology:

$$x(a) = \frac{1}{\beta^\beta (1-\beta)^{1-\beta}} l(a)^\beta \left(\int x(a', a)^{\frac{\sigma-1}{\sigma}} m(a', a) da' \right)^{\frac{\sigma}{\sigma-1} (1-\beta)} \quad (44)$$

Market clearing for intermediate goods:

$$x(a) = \int x(a, a') m(a, a') da' + y(a) \quad (45)$$

Labor market clearing:

$$\int l(a) f(a) da = 1 \quad (46)$$

Laws of motion for product distribution $f(a)$:

$$\frac{\partial f}{\partial t} = -\frac{\partial f}{\partial a} + (\lambda(a) - \delta_F) f(a) + \lambda_E \delta(a) \quad (47)$$

Laws of motion for match distribution $m(a, a')$ where a is supplier and a' is buyer:

$$\frac{\partial m}{\partial t} = -\frac{\partial m}{\partial a} - \frac{\partial m}{\partial a'} + (\lambda(a) - \delta_F - \delta_M) m(a, a') + \frac{\zeta}{N_f} f(a') \quad (48)$$

with boundary condition $m(0, a') = \frac{\zeta_0}{N_f} f(a')$.

R&D labor constraint:

$$L_H = \int \phi(\lambda(a)) f(a) da \quad (49)$$

Index Convention

Throughout this appendix, we use the convention:

- $m(a, a')$: mass of matches where a is the **supplier** age and a' is the **buyer** age
- $V^M(a, a')$: social value of a match where a is supplier and a' is buyer
- $x(a, a')$: quantity supplied from supplier a to buyer a'

Hamiltonian

Define the current-value Hamiltonian for the stationary problem:

$$\begin{aligned}
\mathcal{H} = & \log Y \\
& + \int \mu_f(a) \left[-\frac{\partial f}{\partial a} + (\lambda(a) - \delta_F)f(a) + \lambda_E \delta(a) \right] da \\
& + \iint \mu_m(a, a') \left[-\frac{\partial m}{\partial a} - \frac{\partial m}{\partial a'} + (\lambda(a) - \delta_F - \delta_M)m(a, a') + \frac{\zeta}{N_f}f(a') \right] da da' \\
& + \mu_\lambda \left[L_H - \int \phi(\lambda(a))f(a) da \right]
\end{aligned} \tag{50}$$

Static First-Order Conditions

Define the shadow value of output from product a :

$$v_y(a) \equiv \frac{\partial \log Y}{\partial y(a)} = \frac{y(a)^{-1/\sigma} f(a)}{\int y(\tilde{a})^{\frac{\sigma-1}{\sigma}} f(\tilde{a}) d\tilde{a}} \tag{51}$$

FOC with respect to $x(a, a')$ (intermediate input from supplier a to buyer a'):

$$v_y(a') \cdot (1 - \beta) \cdot x(a') \cdot \frac{x(a, a')^{-1/\sigma}}{\int x(a'', a')^{\frac{\sigma-1}{\sigma}} m(a'', a') da''} = v_y(a) \cdot m(a, a') \tag{52}$$

This equates the marginal benefit to buyer a' (LHS) with the marginal cost to supplier a (RHS).

FOC with respect to $l(a)$ (production labor):

$$v_y(a) \cdot \beta \cdot x(a) = v_L \cdot l(a) \cdot f(a) \tag{53}$$

where v_L is the shadow value of production labor.

Dynamic First-Order Conditions

FOC with respect to $m(a, a')$

The first-order condition with respect to the match distribution $m(a, a')$ gives:

$$\rho \mu_m(a, a') = \frac{\partial \mu_m}{\partial a} + \frac{\partial \mu_m}{\partial a'} + \mu_m(a, a')(\lambda(a) - \delta_F - \delta_M)$$

$$+ \underbrace{v_y(a') \cdot (1 - \beta) \cdot x(a') \cdot \Omega(a, a')}_{\text{Love of variety benefit for buyer } a'} - \underbrace{v_y(a) \cdot x(a, a')}_{\text{Resource cost for supplier } a} \quad (54)$$

where the expenditure share is:

$$\Omega(a, a') \equiv \frac{x(a, a')^{\frac{\sigma-1}{\sigma}}}{\int x(a'', a')^{\frac{\sigma-1}{\sigma}} m(a'', a') da''} \quad (55)$$

The two terms in the flow value have clear economic interpretations:

- **Term 1** (Love of variety): When buyer a' has access to supplier a 's variety, buyer a' 's production efficiency improves. This is the CES love of variety effect.
- **Term 2** (Resource cost): Supplier a uses resources to produce $x(a, a')$ for buyer a' .

Simplification using CES optimality

From the static FOC (52), we have:

$$v_y(a') \cdot (1 - \beta) \cdot x(a') \cdot \Omega(a, a') \cdot x(a, a')^{-1} = v_y(a) \cdot m(a, a') \cdot x(a, a')^{-1} \quad (56)$$

This implies that at the optimum, the love of variety benefit and resource cost are linked through the CES structure. The net benefit simplifies to:

$$\text{Flow value} = (1 - \beta) \cdot r^{SP}(a') \cdot \Omega(a, a') \quad (57)$$

where $r^{SP}(a') = v_y(a') \cdot x(a')$ is the social revenue of buyer a' .

FOC with respect to $f(a)$

$$\begin{aligned} \rho \mu_f(a) &= \frac{\partial \mu_f}{\partial a} + \mu_f(a)(\lambda(a) - \delta_F) \\ &\quad + v_y(a) \cdot y(a) - \mu_\lambda \phi(\lambda(a)) + \frac{\zeta}{N_f} \int \mu_m(a, a') da' \end{aligned} \quad (58)$$

FOC with respect to $\lambda(a)$

$$0 = \mu_f(a)f(a) + \int \mu_m(a, a')m(a, a') da' - \mu_\lambda \phi'(\lambda(a))f(a) \quad (59)$$

Value Function Definitions

Define:

$$V^{SP}(a) \equiv \mu_f(a) + \frac{\int \mu_m(a, a') m(a, a') da'}{f(a)} \quad (60)$$

$$V^M(a, a') \equiv \mu_m(a, a') \quad (61)$$

$$w_H^{SP} \equiv \mu_\lambda \quad (62)$$

Note the index convention: $V^M(a, a')$ is the value of a match where a is supplier and a' is buyer.

Hamilton-Jacobi-Bellman Equations

Product Value Function

Combining the FOCs and rearranging:

$$\begin{aligned} (\rho + \delta_F - \lambda(a))V^{SP}(a) &= r^{SP}(a) + V_a^{SP}(a) - w_H^{SP} \phi(\lambda(a)) \\ &\quad + \int V^M(a, a') \left[\frac{\zeta}{N_f} f(a') - \delta_M m(a, a') \right] da' \end{aligned} \quad (63)$$

where $r^{SP}(a) = v_y(a) \cdot y(a)$ is the social revenue.

The last term is the **network-formation externality wedge**:

$$\text{wedge}(a) = \int V^M(a, a') \left[\frac{\zeta}{N_f} f(a') - \delta_M m(a, a') \right] da' \quad (64)$$

This captures:

- **Formation term** $\frac{\zeta}{N_f} f(a')$: Rate at which supplier a forms new matches with buyers of age a'
- **Destruction term** $\delta_M m(a, a')$: Rate at which existing matches dissolve

Match Value Function

From (54):

$$\begin{aligned} (\rho + \delta_F + \delta_M - \lambda(a))V^M(a, a') &= V_a^M(a, a') + V_{a'}^M(a, a') \\ &\quad + (1 - \beta) \cdot r^{SP}(a') \cdot \Omega(a, a') \end{aligned} \quad (65)$$

The flow value $(1 - \beta) \cdot r^{SP}(a') \cdot \Omega(a, a')$ is the net love of variety benefit. It depends on:

- Buyer a' 's social revenue $r^{SP}(a')$: larger buyers derive more benefit
- Supplier a 's expenditure share $\Omega(a, a')$: suppliers with larger shares contribute more

Optimal R&D Allocation

From the FOC with respect to $\lambda(a)$:

$$\lambda(a) = \left(\frac{\phi}{\gamma w_H^{SP}} V^{SP}(a) \right)^{\frac{1}{\gamma-1}} \quad (66)$$

This gives R&D labor demand:

$$l_H^{SP}(a) \propto \left(V^{SP}(a) \right)^{\frac{\gamma}{\gamma-1}} \quad (67)$$

Comparison with Decentralized Equilibrium

In the decentralized equilibrium (Proposition 2):

$$(\rho + \delta_F - \lambda(a))V(a) = \pi(a) + V_a(a) - w_H \phi(\lambda(a)) \quad (68)$$

where $\pi(a) = (1 - 1/\mu)r(a)$ is private profit.

Two differences from the Social Planner:

1. **Market power distortion:** $\pi(a)$ vs $r^{SP}(a)$. Markups suppress private returns.
2. **Network-formation externality:** The wedge term is absent in the decentralized problem. Firms do not internalize love of variety benefits for trading partners.

E Constrained Planner's Problem

The Constrained Planner maximizes welfare subject to maintaining the decentralized goods market structure. This isolates the network-formation externality from market power distortions.

Setup

The Constrained Planner solves:

$$\max \int_0^\infty e^{-\rho t} \log Y(t) dt \quad (69)$$

subject to:

- Decentralized static allocation: $c(a)$, $D(a)$, $r(a)$ from equations (9) and (8)
- Laws of motion for $f(a)$ and $m(a, a')$
- R&D labor constraint

The key difference from the Social Planner: the static allocation is **not** optimized. Markups remain.

Hamiltonian

$$\begin{aligned} \mathcal{H}^{CP} = & \log Y^{CP} \\ & + \int \mu_f^{CP}(a) \left[-\frac{\partial f}{\partial a} + (\lambda(a) - \delta_F)f(a) + \lambda_E \delta(a) \right] da \\ & + \iint \mu_m^{CP}(a, a') \left[-\frac{\partial m}{\partial a} - \frac{\partial m}{\partial a'} + (\lambda(a) - \delta_F - \delta_M)m(a, a') + \frac{\zeta}{N_f} f(a') \right] da da' \\ & + \mu_\lambda^{CP} \left[L_H - \int \phi(\lambda(a))f(a) da \right] \end{aligned} \quad (70)$$

where Y^{CP} is computed using the decentralized static allocation.

Key Insight

Since the static allocation is fixed at the decentralized level, the FOCs with respect to $f(a)$ and $m(a, a')$ yield the same structure as the Social Planner, but with decentralized revenues $r(a)$ instead of social revenues $r^{SP}(a)$.

Hamilton-Jacobi-Bellman Equations

Product Value Function

$$(\rho + \delta_F - \lambda^{CP}(a))V^{CP}(a) = r(a) + V_a^{CP}(a) - w_H^{CP} \phi(\lambda^{CP}(a))$$

$$+ \int V^{M,CP}(a, a') \left[\frac{\zeta}{N_f} f(a') - \delta_M m(a, a') \right] da' \quad (71)$$

Note: The flow return is $r(a)$ (decentralized revenue with markups), not $r^{SP}(a)$.

Match Value Function

$$(\rho + \delta_F + \delta_M - \lambda^{CP}(a))V^{M,CP}(a, a') = V_a^{M,CP}(a, a') + V_{a'}^{M,CP}(a, a') + (1 - \beta) \cdot r(a') \cdot \Omega(a, a') \quad (72)$$

The flow value uses decentralized revenue $r(a')$.

Optimal R&D Allocation

$$I_H^{CP}(a) \propto \left(V^{CP}(a) \right)^{\frac{\gamma}{\gamma-1}} \quad (73)$$

Comparison of Three Allocations

	Decentralized	Constrained Planner	Social Planner
Flow return	$\pi(a) = (1 - 1/\mu)r(a)$	$r(a)$	$r^{SP}(a)$
Wedge term	No	Yes	Yes
Static allocation	Decentralized	Decentralized	Optimized

The Constrained Planner differs from Decentralized by adding the wedge term. The Social Planner additionally optimizes the static allocation, replacing $r(a)$ with $r^{SP}(a)$.

F Proof of Proposition 4

We prove that in partial equilibrium, an increase in the match measure $m(a, a')$ with supplier a raises buyer a' 's optimal innovation rate $\lambda(a')$.

Setup. Fix the aggregate price index P , R&D wage w_H , suppliers' cost shifters $\{c(a)\}_{a \geq 0}$, buyer's demand shifter $D(a')$, and continuation value $V_{a'}(a')$.

The buyer's equilibrium is characterized by four equations:

Cost shifter:

$$c(a')^{1-\sigma} = \left(\int c(\tilde{a})^{1-\sigma} m(\tilde{a}, a') d\tilde{a} \right)^{1-\beta} \quad (C.1)$$

Revenue:

$$r(a') = \left(\frac{\mu c(a')}{P} \right)^{1-\sigma} D(a') \quad (\text{C.2})$$

HJB equation:

$$(\rho + \delta_F - \lambda(a'))V(a') = (1 - 1/\mu)r(a') + V_{a'}(a') - w_H \phi \lambda(a')^\gamma \quad (\text{C.3})$$

R&D first-order condition:

$$w_H \gamma \phi \lambda(a')^{\gamma-1} = V(a') \quad (\text{C.4})$$

Step 1: Effect of $m(a, a')$ on $c(a')$.

Define effective supplier variety:

$$M(a') \equiv \int c(\tilde{a})^{1-\sigma} m(\tilde{a}, a') d\tilde{a}$$

From (C.1):

$$\frac{\partial c(a')^{1-\sigma}}{\partial m(a, a')} = (1 - \beta)M(a')^{-\beta} \cdot c(a')^{1-\sigma} > 0$$

Since $\sigma > 1$, an increase in $c(a')^{1-\sigma}$ implies a decrease in $c(a')$:

$$\frac{\partial c(a')}{\partial m(a, a')} < 0$$

Step 2: Effect of $m(a, a')$ on $r(a')$.

From (C.2):

$$r(a') = \mu^{1-\sigma} P^{\sigma-1} D(a') \cdot c(a')^{1-\sigma}$$

Thus:

$$\frac{\partial r(a')}{\partial m(a, a')} = \mu^{1-\sigma} P^{\sigma-1} D(a') \cdot (1 - \beta)M(a')^{-\beta} c(a')^{1-\sigma} > 0$$

Step 3: Effect of $r(a')$ on $\lambda(a')$.

Substitute (C.4) into (C.3):

$$(\rho + \delta_F - \lambda)w_H \gamma \phi \lambda^{\gamma-1} = (1 - 1/\mu)r + V_{a'} - w_H \phi \lambda^\gamma$$

Rearranging:

$$\Phi(\lambda; r) \equiv w_H \phi \lambda^{\gamma-1} [\gamma(\rho + \delta_F) + (1 - \gamma)\lambda] - (1 - 1/\mu)r - V_{a'} = 0$$

By implicit differentiation:

$$\frac{d\lambda}{dr} = \frac{1 - 1/\mu}{\partial\Phi/\partial\lambda}$$

where:

$$\frac{\partial\Phi}{\partial\lambda} = w_H\phi(\gamma - 1)\lambda^{\gamma-2} [\gamma(\rho + \delta_F) - \lambda]$$

Since $\gamma > 1$ and $\rho + \delta_F > \lambda$ (required for the HJB to have a well-defined solution), we have $\frac{\partial\Phi}{\partial\lambda} > 0$. Therefore:

$$\frac{d\lambda}{dr} > 0$$

Step 4: Combining.

$$\frac{d\lambda(a')}{dm(a, a')} = \frac{d\lambda}{dr} \cdot \frac{\partial r}{\partial m(a, a')} > 0$$

An increase in the match measure $m(a, a')$ raises the buyer's optimal innovation rate $\lambda(a')$. ■

G Numerical Appendix

Product Distribution f

We solve the model with finite difference methods. Throughout this section, to construct the derivative matrices, we use a backward approximation when the drift of the state variable is positive, and a forward approximation when the drift of the state is negative. Notice that the stationary distribution is the solution for the following differential equation:

$$0 = -\frac{\partial f(a)}{\partial a} + (\lambda(a) - \delta_F) f(a) + \lambda_E \delta(a) \quad (74)$$

We now discretize a on an evenly spaced $N_a \times 1$. Let D_a be the $N_a \times N_a$ matrix that, when pre-multiplying f , gives an approximation of f_a . Analogously, define D_a :

$$f_a = D_a f$$

Vectorize (74) and obtain,

$$f = -\{-D_a + \lambda - \delta_F\}^{-1} \lambda_E \delta$$

where the element of N_a vector f consists of $f(a)$, the element of N_a vector λ consists of $\lambda(a)$, and the element of N_a vector δ consists of $\delta(a)$

Matched Product Distribution m

The distributions of matched products $m(a', a)$ is given by

$$\frac{\partial}{\partial a} m(a', a) = -\frac{\partial}{\partial a'} m(a', a) + (\lambda(a') - \delta_F - \delta_M) m(a', a) + \frac{\zeta_0 \lambda_E}{N} f(a') + \zeta_0 \frac{\lambda_E}{N_f} \delta(a')$$

subject to the boundary condition $m(a'; 0) = 0$. Vectorize this, and obtain

$$\begin{aligned} \frac{m - m_{-da}}{da} &= (-D_a + \lambda(a') - \delta_F - \delta_M) m + \frac{\zeta_0 \lambda_E}{N} f + \zeta_0 \frac{\lambda_E}{N_f} \delta \\ m - m_{-da} &= da (-D_a + \lambda(a') - \delta_F - \delta_M) m + da \frac{\zeta_0 \lambda_E}{N} f + da \zeta_0 \frac{\lambda_E}{N_f} \delta \\ m &= \{I_a - da (-D_a + \lambda(a') - \delta_F - \delta_M)\}^{-1} \left(m_{-da} + da \frac{\zeta_0 \lambda_E}{N} f + da \zeta_0 \frac{\lambda_E}{N_f} \delta \right) \end{aligned}$$

Starting from $m(\cdot; 0) = 0$, the forward iteration of the above vectorized equation gives the distributions of matched products $m(a', a)$ for each a . Note that $m(a', a)$ converges as a becomes sufficiently large. Therefore, we only need to forward iterate until $m(a', a)$ converges.

Value Function V

Let Δ denote step-size and τ the iteration of the algorithm. Then given $V^{\tau-1}(a)$, the implicit method gives an update

$$\frac{1}{\Delta} (V^\tau(a) - V^{\tau-1}(a)) + (\rho + \delta_F) V^\tau(a) = \pi(a) + V_a^\tau(a) + \lambda(a) V^\tau(a) - w_H \phi(\lambda(a))$$

where $\pi(a) = \left(1 - \frac{1}{\mu}\right) \left(\frac{\mu c(a)}{P}\right)^{1-\sigma} D(a)$. Rearranging this,

$$\left(\frac{1}{\Delta} + \rho + \delta_F - \lambda(a)\right) V^\tau(a) - V_a^\tau(a) = \pi(a) - w_H \phi(\lambda(a)) + \frac{1}{\Delta} V^{\tau-1}(a)$$

Now, we vectorize the HJB equation:

$$\left(\frac{1}{\Delta} + \rho + \delta_F - \Lambda - D_a\right) V^\tau = \pi - w_H \phi(\lambda) + \frac{1}{\Delta} V^{\tau-1}$$

$$V^\tau = \left\{ \frac{1}{\Delta} + \rho + \delta_F - \Lambda - D_a \right\}^{-1} \left(\pi - w_H \phi(\lambda) + \frac{1}{\Delta} V^{\tau-1} \right)$$

where the element of the N_a -dimensional vector V^τ consists of $V^\tau(a)$, the element of the N_a -dimensional vector π consists of $\pi(a)$, $\Lambda \equiv \text{diag}(\lambda(a_i))$ collects the age-specific innovation rates along the grid $\{a_i\}_{i=1}^{N_a}$, and $\phi(\lambda)$ denotes the vector whose entries are $\phi(\lambda(a_i))$. The implicit method works by updating V^τ through the above equation.

Social Matching Value Function V^M

$$(\rho + \delta_F + \delta_M - \lambda(a)) V^M(a, a') = \frac{\partial}{\partial a} V^M(a, a') + \frac{\partial}{\partial a'} V^M(a, a') + (1 - \beta) r^{SP}(a') \Omega(a, a')$$

We now discretize a and a' on evenly spaced grids with N_a nodes each. Stack these according to:

$$\begin{pmatrix} a_1, a'_1 \\ a_2, a'_1 \\ \vdots \\ a_{N_a}, a'_1 \\ \vdots \\ a_1, a'_{N_a} \\ \vdots \\ a_{N_a}, a'_{N_a} \end{pmatrix}$$

$$(\rho + \delta_F + \delta_M - \Lambda_a - D_a - D_{a'}) V^M = A$$

$$V^M = (\rho + \delta_F + \delta_M - \Lambda_a - D_a - D_{a'})^{-1} A$$

where the N_a^2 -dimensional vector V^M collects $V^M(a, a')$ evaluated on the stacked grid, $\Lambda_a \equiv I_{N_a} \otimes \Lambda$ applies the age-specific innovation rates to the first argument, D_a and $D_{a'}$ are the finite-difference operators with respect to a and a' , and the N_a^2 -dimensional vector A has elements

$$A = (1 - \beta) r^{SP}(a') \Omega(a, a') \quad \text{for each stacked pair } (a, a').$$